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Unlabeled Multi-Robot Motion Planning with Improved Separation Trade-offs

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Abstract

We study unlabeled multi-robot motion planning for unit-disk robots in a polygonal environment. In this problem, a set of indistinguishable robots must navigate from a set of starting positions to an equal number of target positions. The objective is to compute continuous, collision-free paths that route every robot to a distinct target, while minimizing the total distance traveled. Although the problem is hard in general, polynomial-time solutions exist under appropriate separation assumptions on the start and target positions. Banyassady et al. (SoCG'22) guarantee feasibility in simple polygons under start–start and target–target distances of at least 4, and start–target distances of at least 3, but without optimality guarantees. Solovey et al. (RSS'15) provide a near-optimal solution in general polygonal domains, where robots move to their targets in a sequential manner (i.e. a monotone motion plan), under stricter conditions: start/target positions must have pairwise distance at least 4, and at least $\sqrt{5} \approx 2.236$ from obstacles. This raises the question of whether polynomial-time algorithms can be obtained in even more densely packed environments.

In this thesis we present a generalized algorithm that achieves different tradeoffs on the robots-separation and obstacles-separation bounds, all significantly improving upon the state of the art. Specifically, we obtain polynomial-time constant-approximation algorithms to minimize the total path length when (i) the robots-separation is $2\frac{2}{3}$ and the obstacles-separation is $1\frac{2}{3}$, or (ii) the robots-separation is ≈ 3.291 and the obstacles-separation ≈ 1.354. These solutions are weakly-monotone (i.e. robots may move slightly in their assigned revolving area during the entire motion plan); we also provide a monotone solution requiring obstacles-separation of ≈ 1.614 and robots-separation of 4. We prove that monotone plans may not exist below obstacles-separation ≈ 1.614, and weakly-monotone plans may not exist below ≈ 1.354. Additionally, we introduce a different strategy yielding a polynomial-time solution when the robots-separation is only 2, and the obstacles-separation is 3. Finally, we show that without any robots-separation assumption, obstacles-separation of at least 1.5 may be necessary for a solution to exist.

The results in this thesis are based on joint work by Tsuri Farhana, Omrit Filtser and Shalev Goldshtein, and were published in the 42nd International Symposium on Computational Geometry (SoCG 2026) [13].

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1 Introduction

In the classic multi-robot motion-planning (MRMP) problem, a set of robots operate in a common workspace, and need to reach a set of target positions. More precisely, given a set of m starting positions and a set of m target positions in a workspace $\mathcal{W}$, the goal is to compute a motion strategy for the robots to reach the targets while avoiding collisions with the boundary of the workspace and with other robots. In this thesis we consider unit-disk robots that are moving in a polygonal workspace in the plane. If each robot is assigned to a specific target, the problem is referred to as *labeled* MRMP, and otherwise it is *unlabeled* MRMP. Unless otherwise specified, the variant of MRMP that we consider in this thesis is unlabeled.

To illustrate the problem setting, Figure 1 depicts a simple instance of the MRMP problem with two robots moving amid polygonal obstacles.

The MRMP problem and its variants have been widely investigated, both from theoretical and practical perspectives; for some of the more recent results, see, for example, [3–6, 12, 14]. Very recently, both the labeled and unlabeled variants of MRMP for unit disks in polygons with holes were shown to be PSPACE-hard [1]. Many other variants of the problem are known to be hard as well, see e.g. [9, 16, 28, 31].

Interestingly, the separation between the robots plays a key role in the difficulty of the problem. Adler, de Berg, Halperin, and Solovey [2] show that in a simple polygonal workspace with n vertices and m unit-disk robots, when the ***robots-separation***, denoted ρ, is 4 (i.e., all the start and target positions are at least distance 4 apart), a solution for unlabeled MRMP always exists, and can be found in $\tilde{O}(mn + m^2)$ time.

In [8] this result is further improved by distinguishing between two types of robots-separation bounds: (i) ***monochromatic*** robots-separation — the distance between two start positions or between two target positions, and (ii) ***bichromatic*** robots-separation — the distance between a start and a target position. They present an algorithm with the same running time

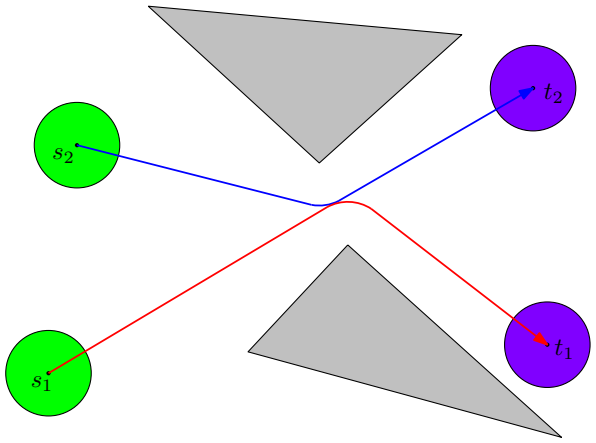


Figure 1: An example of a small MRMP instance. Throughout this thesis, green disks indicate starting positions (s_i) and purple disks indicate target positions (t_i). The workspace contains multiple polygonal obstacles (gray). The paths (red and blue curves) represent collision-free trajectories for the robots. Notice that while the two paths geometrically intersect in the narrow passage, a valid motion plan coordinates the movement such that the robots do not occupy the intersection point simultaneously, thereby avoiding a physical collision.

of $O(n \log n + mn + m^2)$, but under weaker assumptions: monochromatic separation of 4, and bichromatic separation of 3. Moreover, they prove that these bounds are tight, i.e., a solution for unlabeled MRMP in a simple polygon may not always exist when the monochromatic separation is $4 - \varepsilon$, or when the bichromatic separation is $3 - \varepsilon$. In addition, they show that when the free space consists of a single connected component, the bichromatic separation is not required for a solution to always exist.

The algorithmic approach in both papers [2, 8] is similar: restricting the robots to move one at a time (a ***monotone*** strategy) on a motion graph that has the start and target positions as vertices. A drawback of this approach is that it does not provide any reasonable approximation on the length of the solution, i.e., the total length of the paths in the motion plan.

Solovey, Yu, Zamir, and Halperin [30] present a different approach for un-

labeled MRMP, with optimality guarantee even in polygons with holes, but at the cost of adding an additional assumption on the distance between the obstacles to any start and target position. Specifically, they consider the case when the robots-separation is $\rho = 4$, and the ***obstacles-separation***, denoted ω, is $\sqrt{5} \approx 2.236$ (i.e., the distance between any start/target position and any obstacle is at least $\sqrt{5}$). To clarify these geometric constraints conceptually, Figure 2 visually defines the parameters ρ and ω. Under these assumptions, they present an algorithm that runs in $\tilde{O}(m^4 + n^2m^2)$ time and returns a solution of total length at most OPT $+ 4m$. Here and throughout the thesis, OPT denotes the minimum possible sum of lengths of the paths in a collision-free motion plan.

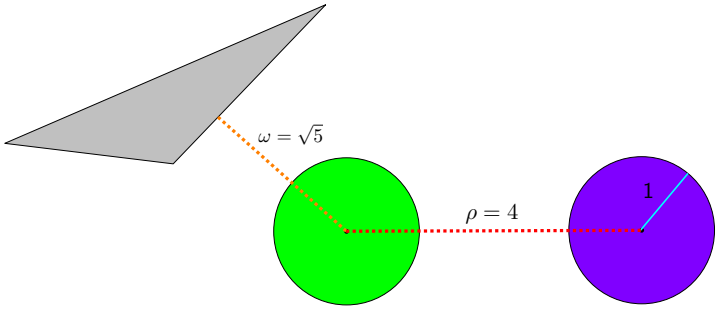


Figure 2: Visual definition of the separation parameters, illustrating the assumptions of $\rho = 4$ and $\omega = \sqrt{5}$ as used by Solovey et al. [30]. The robots-separation ρ represents the required minimal distance between any two positions (start–start, target–target, or start–target) on different robots, dictating the minimum center-to-center distance. The obstacles-separation ω represents the required minimal clearance from the center of a robot (at either its start or target position) to the boundary of any environmental obstacle (shown in gray). The unit radius is indicated by the cyan line.

Notice that when assuming a large robots-separation ($\rho \geq 4$), the overall "density" of the robots inside the workspace becomes small, which may require the workspace to be much larger than the number of robots operating in it. Also, assuming a large obstacles-separation requires adding

an additional buffering "layer" around the obstacles. This motivates the question of whether we can find weaker assumptions on either the robots-separation or the obstacles-separation, that are sufficient for having a polynomial time algorithm for unlabeled MRMP. Surprisingly, in this thesis we show that both the robots-separation and obstacles-separation assumptions can be significantly relaxed, while still enabling a polynomial-time algorithm that achieves a constant-factor approximation of the solution length.

In previous works, namely [2, 8, 30], the algorithmic approach uses a ***monotone*** strategy, where robots move to their target position one by one according to some order, while other robots must remain at their start/target positions. The main result in our thesis uses a weakly-monotone strategy. In a ***weakly-monotone*** motion plan, each start/target position p is associated with a revolving area - a disk of radius $r > 1$ that contains the unit disk centered at p and is empty of obstacles and other unit disks centered at other start/target positions. The robots move to their target position one by one according to some order, while other robots must remain inside the revolving area associated with their current position. Such a weakly-monotone strategy was used, e.g., in [4, 27] for a variant of labeled MRMP with unit-disk robots.

1.1 Contribution of this thesis

In this thesis we present two different algorithms for unlabeled MRMP, that provide tighter trade-offs between robot-separation and obstacles-separation assumptions. In particular, for a parameter $0 \leq \varepsilon \leq 1$, a polygonal environment of complexity n, and m unit-disc robots, our first algorithm computes in $\tilde{O}(m^4 + n^2m^2)$ time a collision-free motion plan of total length $O(\mathrm{OPT})$, assuming that $\rho = \max\{4 - 2\varepsilon, 2 + \varepsilon\}$ and $\omega = \max\{1 + \varepsilon, \sqrt{(3 - \sqrt{3} - \varepsilon)^2 + 1}\}$ (see Theorem 14). In Table 1.1 we highlight three interesting values of ε that minimize either ρ, ω or the guaranteed length of the solution. This algorithm can be viewed as a generalization of the algorithm in [30], and indeed, as a consequence of our analysis we positively resolve a conjecture made in [30], by showing that $\omega = 1.614$ is sufficient when $\rho = 4$. Moreover, in Chapter 6 we show that a monotone motion plan (such as the one produced by the algorithm in [30]) may not always exist when $\omega < 1.614$, and therefore to achieve smaller separation bounds one must use a different strategy. Indeed, as mentioned above,

our motion plan is weakly-monotone. Interestingly, we also show that a weakly-monotone motion plan may not always exist when $\omega < 1.354$.

Note that our first algorithm does not cover the case of $\rho = 2$, which is the optimal (monochromatic) robots-separation. We therefore present a second algorithm using a completely different approach, which given a simple polygon with n vertices and m unit-disk robots, computes in $O(m^3 + mn^2)$ time a collision-free motion plan of total length OPT $+ O(m^2)$, assuming that $\rho = 2$ and $\omega = 3$ (see Theorem 16).

Finally, we present lower bounds on the amount of obstacles-separation that is required in order for a solution to always exist. In the unlabeled case, we show that when no assumptions are made on ρ (i.e., $\rho = 2$), obstacles-separation of at least 1.5 is sometimes needed. In the labeled case, we observe that obstacles-separation of at least 2 is sometimes necessary.

ω	ρ	**Solution length**	**Running time**	**Reference**
1	4	no guarantee	$\tilde{O}(mn + m^2)$	[2]*
1	4, $(\beta = 3)$	no guarantee	$\tilde{O}(mn + m^2)$	[8]*
$\sqrt{5}$	4	$OPT + 4m$	$\tilde{O}(m^4 + n^2m^2)$	[30]
≈ 1.614	4	$OPT + 4m$	$\tilde{O}(m^4 + n^2m^2)$	Theorem 14
≈ 1.354	≈ 3.291	$O(OPT)$	$\tilde{O}(m^4 + n^2m^2)$	Theorem 14
$1\frac{2}{3}$	$2\frac{2}{3}$	$O(OPT)$	$\tilde{O}(m^4 + n^2m^2)$	Theorem 14
3	2	$OPT + O(m^2)$	$\tilde{O}(m^3 + mn^2)$	Theorem 16*

Table 1.1: Summary of algorithms and their separation assumptions. Results marked with an asterisk (*) apply only to simple polygons.

1.2 Technical ideas and overview

Our approach in the first algorithm generalizes the strategy of Solovey et al. [30] for unlabeled MRMP, which is briefly described as follows. Given a set S of start positions, a set T of target positions, and a polygonal workspace $\mathcal{W}$, first compute an optimal set Γ of paths assigning each starting point to a target point (a perfect matching). This set of paths minimizes the sum

of geodesic distances, when ignoring the other robots in the workspace. Then, one can show that when assuming $\rho = 4$ and $\omega = \sqrt{5}$, there is always at least one target which is a so-called standalone goal, that is, a target t that does not block any other path in Γ. Such a target is useful because we can place a robot there and then treat it as an obstacle. However, there might be some other robot that blocks the path that leads to t, in which case we can make a switch: we push the last robot that blocks this path to t. This switching process is possible because of the assumption $\omega = \sqrt{5}$. Then, run the algorithm recursively, but after removing the start and target positions that were satisfied in the previous recursive step, and updating the workspace to treat the robot at t as an obstacle.

Our weakly-monotone motion plan is obtained by inserting a very important modification to the algorithm of [30]: instead of choosing a standalone target in each iteration, we relax the requirement and require only an "almost standalone" target — a target that may interrupt other paths, but only by a small amount. In our strategy, we distinguish between "blocking" and "interrupting" positions. Instead of always switching paths when another robot is standing in the way, we allow robots that interrupt a path by only a small amount to move away from the path. A robot that interrupts a given path may move slightly in its close neighborhood (a disk of radius equal to the amount of interruption plus one) in order to clear the way. In contrast, a robot that blocks a path may not have enough wiggle room, and we will switch the paths.

As suggested above, this combined strategy can be viewed as a generalization of [30]: we introduce another parameter, called the "overlap parameter", which is used to determine whether a position is blocking a path or only interrupting it. This parameter allows us to define a set of constraints on ρ and ω, that allow (i) the existence of an interrupting target, (ii) the option to switch paths in the case when a robot is blocking a path, and (iii) the existence of a sufficiently large free neighborhood for an interrupting robot to clear the way. It turns out that there is an interesting trade-off between these constraints, leading to the main result of our thesis (Theorem 14).

Note that the concept of interrupting positions raises some issues that were not present in the algorithm of [30]. For example, in each iteration of the recursive algorithm we need to take into account that the robot placed on a target may still interrupt other paths. In [30] this is solved simply by removing a unit disk around the interrupting target from the workspace. However, in our case, such a robot may still need to move in order to clear

the way for other robots. Moreover, if we simply treat it as an obstacle, the assignment of paths in the next iteration may be too large. We solve this issue by observing that it is actually enough to remove a smaller disk around a settled target when computing the paths, while still allowing it to move within its small neighborhood when planning the motion in the following iterations. Another issue is that there may be two or more robots that interrupt a path and may need to move simultaneously. Here we describe a simple motion with respect to the ray connecting the robot moving on the path with the interrupting robots.

In Chapter 4, we systematically define and analyze the constraints that are required for each part of this strategy, each constraint depends on the overlap parameter. In Chapter 5 we present the complete generalized algorithm and prove its correctness. Then, we show how to choose an overlap parameter and adjust constraints to get the minimum possible obstacles-separation and the minimum possible robots-separation.

In our first algorithm, to clear a path, we only move robots that intersect that path. Such a strategy cannot work when the robots-separation is exactly 2, because a robot that intersects a path may be blocked by other robots that do not intersect that path, and thus will not be able to move and clear the path. Therefore, to achieve a separation bound of 2 between the robots, we present in Chapter 7 a completely different strategy, which we call the Exodus algorithm. In this strategy, we also iteratively pick a path between start and target positions that were not yet satisfied, and move a robot along that path, however, in contrast to the first algorithm, we clear the path completely by also moving robots that do not intersect the path. In simple polygons, this strategy allows us to assume robots-separation of only 2, with obstacles-separation of 3, and it applies also for labeled MRMP (see Theorem 16).

Note that the monochromatic separation must be at least 2, so the robots-separation is almost optimal (the bichromatic separation can be 0). In Chapter 9, we preset lower bounds on the obstacles-separation, and we leave open the question of what is the upper bound on ω when robots-separation is optimal.

2 Background and Previous Work

The theoretical foundations of Multi-Robot Motion Planning were established with the "Piano Movers"' problem series by Schwartz and Sharir in 1983 [23]. For the labeled variant, they demonstrated that the motion of two or three disk robots in a 2D polygonal workspace could be coordinated in $O(n^3)$ and $O(n^{13})$ time, respectively [24]. For the unlabeled variant, the coordination of multiple indistinguishable disks can be solved exactly using algebraic techniques, though the running time scales exponentially with the number of robots. These early results were later refined by Yap [35] using a retraction-based approach to achieve $O(n^2)$ time for two robots, and by Sharir and Sifrony [25] using cell decomposition.

Despite successes for a constant number of robots—where polynomial-time exact or constant-factor approximation solutions are known (see, e.g., [3, 5, 6, 12])—the general problem remains computationally intractable when the number of robots m is part of the input.

2.1 Computational Complexity and Hardness

The complexity of MRMP heavily depends on whether the robots are labeled or unlabeled, and on the specific geometry of the robots and the workspace.

Early work by Spirakis and Yap [31] established that planning for multiple labeled disk robots moving amid polygonal obstacles is strongly NP-hard. Hopcroft et al. [16] proved PSPACE-hardness for coordinating the motion of multiple labeled rectangles. Since these early ad-hoc proofs were notoriously complex, Hearn and Demaine [15] later introduced the Nondeterministic Constraint Logic (NCL) framework to unify and simplify hardness proofs for multi-agent systems. By modeling continuous robot motions as discrete edge-reversals in a weighted graph, NCL provides a pow-

erful reduction tool that has since become the standard for analyzing robot coordination complexity.

For a long time, the unlabeled variant was conjectured to be easier due to the flexibility in target assignments. However, Solovey and Halperin [28] utilized this NCL framework to prove that unlabeled MRMP is indeed PSPACE-hard when considering unit-square robots moving amid polygonal obstacles. Hardness was also shown for unlabeled k-colored disks [9]—a partially-labeled variant where robots are partitioned into k distinct color classes, and any robot can occupy any target of its own color.

Most recently, Abrahamsen et al. [1] definitively closed the gap for the standard setting, proving PSPACE-hardness for the reconfiguration of unit disks in polygons with holes. This recent result implies PSPACE-hardness for both the labeled and unlabeled variants of unit disks in environments with holes, leaving open only very specific structural edge cases (such as unit disks in strictly simple polygons without separation assumptions).

2.2 Separation Bounds and Monotone Strategies

This inherent intractability motivates the search for restricted, yet meaningful, structural settings that allow for polynomial-time planning. Adler et al. [2] demonstrated that for unlabeled unit discs in simple polygons, a solution always exists if the general robots-separation ρ is at least 4. Their algorithm works by defining a discrete motion graph over the start and target positions, computing a valid sequence of node-to-node transfers, and then translating these discrete graph moves into continuous workspace motions using a *monotone* strategy. In a monotone strategy, robots move sequentially one at a time, while all other passive robots remain stationary at their designated start or target positions.

Banyassady et al. [8] refined this condition by distinguishing two types of robots-separation bounds: (i) ***monochromatic*** robots-separation (μ) — the distance between two start positions or between two target positions, and (ii) ***bichromatic*** robots-separation (β) — the distance between a start and a target position. They established that $\mu = 4$ and $\beta = 3$ guarantee a solution in $\tilde{O}(mn + m^2)$ time. They also proved that these bounds are tight: a solution in a simple polygon may not exist when $\mu = 4 - \varepsilon$ or $\beta = 3 - \varepsilon$. Additionally, they observed that when the free space consists of a single connected component, the bichromatic separation is not strictly required for a solution to exist.

However, a major drawback of these monotone strategies is that they prioritize feasibility over optimality. By moving robots sequentially on predefined graphs, the resulting motion plans offer no reasonable approximation guarantees on the total geodesic path length. As discussed in detail in the introduction (Chapter 1), Solovey et al. [30] achieved optimal-length guarantees by introducing an *obstacles-separation* parameter ($\omega = \sqrt{5}$), allowing them to efficiently route robots in general polygons with holes.

2.3 The Revolving Areas Model and Weakly-Monotone Plans

When optimality is desired without strict global distance constraints, researchers have explored localized sparsity models. For labeled MRMP, Agarwal, Geft, Halperin, and Taylor [4] (following Solomon and Halperin [27]) consider unit disk robots with ***revolving areas***. In this model, each start and each target position is contained within a bounding disk of radius 2 lying completely in the free workspace. This bounding disk is not necessarily concentric with the start or target position, but it is strictly free from other start or target positions.

Interestingly, although under this assumption we may have problem instances with extreme global density parameters (e.g., monochromatic separation $\mu = 2$, bichromatic separation $\beta = 0$, and obstacles-separation $\omega = 1$), the overall density of the robots inside the workspace remains fundamentally small due to the localized revolving areas. This local sparsity allows for a solution to always exist. Agarwal et al. [4] presented a motion strategy which is *weakly-monotone*: robots move to their target according to some ordering, one at a time, while other passive robots may only maneuver locally within their respective revolving areas to yield the right-of-way. The total length of the solution returned by their algorithm is a constant approximation of the optimal strategy.

2.4 Sampling-Based Techniques

To contrast geometric approaches such as ours, it is useful to briefly review sampling-based (SB) techniques. When strong geometric structure is absent, motion planning is often addressed using these methods, which avoid an explicit construction of the configuration space. Instead, they ex-

plore it implicitly by sampling feasible configurations and attempting to connect nearby samples. A classical example is the Probabilistic Roadmap (PRM) method [17], which constructs a reusable roadmap suitable for multi-query settings. For single-query problems, RRT-Connect [20] incrementally grows trees from the start and target configurations and attempts to connect them efficiently.

While SB methods are highly effective in high-dimensional spaces, their direct application to multi-robot systems is challenging. The joint configuration space grows linearly in dimension with the number of robots, while the size of the space to be explored grows exponentially with the total number of degrees of freedom [22]. As a result, sampling-based exploration may become impractical in dense multi-robot settings, even when probabilistic guarantees hold asymptotically.

To address this issue, specialized algorithms such as discrete RRT (dRRT) [29] and its asymptotically optimal variant dRRT* [26] were proposed. These methods operate over implicit tensor-product roadmaps, combining sampling with graph-search techniques to explore the composite configuration space more efficiently. Nevertheless, their guarantees are asymptotic and probabilistic in nature, rather than deterministic worst-case bounds on running time.

2.5 Discrete Paradigms

Another complementary perspective arises from discrete formulations. An alternative approach to MRMP is to discretize the problem into graph-based models such as pebble motion [7, 19], where robots move between vertices and collisions are defined combinatorially. This abstraction forms the basis of the Multi-Agent Path Finding (MAPF) framework [32], where agents traverse a graph while avoiding vertex and edge conflicts.

MAPF has been extensively studied, with objectives such as makespan and sum-of-costs. However, optimal MAPF is computationally hard even on restricted graph classes [36, 37]. Despite this, the discrete setting has led to a wide range of exact, bounded-suboptimal, and approximation algorithms, as well as insights into coordination in dense environments [11, 14].

Although these discrete models provide useful algorithmic tools, they abstract away the underlying geometry of the workspace. In particular, they do not directly capture geometric clearance constraints or continuous colli-

sion avoidance, which are central to the separation-based guarantees studied in this thesis. As a result, while discrete approaches offer valuable intuition and heuristics, their guarantees apply to the discrete abstraction and do not immediately imply the continuous geometric guarantees that are the focus of our work.

3 Preliminaries

We consider a set of m unit-disk robots moving in a planar polygonal ***workspace***, denoted by $\mathcal{W} \subseteq \mathbb{R}^2$. The workspace may be a simple polygon or cluttered with polygonal obstacles, and we denote its overall number of edges by n.

The ***obstacle space***, denoted by $\mathcal{O}$, is defined as the complement of the workspace in the plane, that is, $\mathcal{O} = \mathbb{R}^2 \setminus \mathcal{W}$. We refer to the points in $\mathcal{W}$ as positions, and we say that a robot is at position $x \in \mathcal{W}$ when its geometric center is located at the point x. For a point $x \in \mathbb{R}^2$ and a radius $r \in \mathbb{R}_+$, we define $D_r(x)$ to be the open disk of radius r centered at x.

Because we model the unit-disk robots as open sets, two robots are considered to collide if and only if the Euclidean distance between their center positions is strictly less than 2. Furthermore, a robot collides with the static environment if its center is at a distance strictly less than 1 from the obstacle space $\mathcal{O}$.

Consequently, we define the ***free space***, denoted by $\mathcal{F}$, as the set of all valid positions in $\mathcal{W}$ where a unit-disk robot does not collide with the obstacle space:

$$\mathcal{F} = \{x \in \mathbb{R}^2 \mid D_1(x) \cap \mathcal{O} = \emptyset\}.$$

In standard motion planning terminology, this corresponds to the free *Configuration Space* (free C-space) of the robot. Geometrically, the collision regions (known as C-space obstacles) are formed by the Minkowski sum of the obstacle space $\mathcal{O}$ and the robot's shape. The free space $\mathcal{F}$ is simply the complement of these C-space obstacles. Throughout this thesis, $\mathcal{F}$ is treated as a closed set, and we denote its boundary by $\partial\mathcal{F}$.

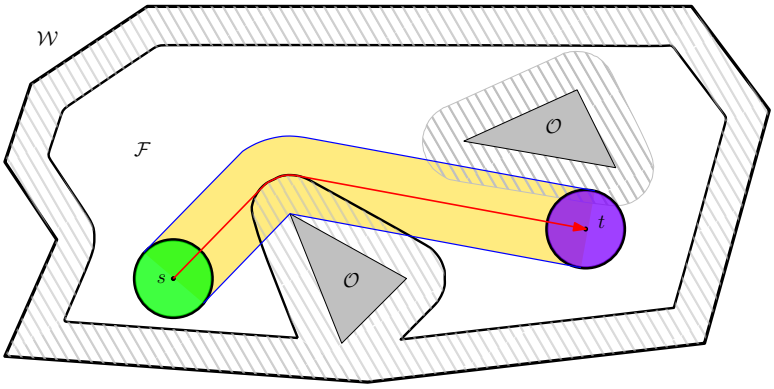


Figure 3: Illustration of the problem geometry and notation. The bounded region defined by the outer polygon is the workspace $\mathcal{W}$. The obstacle space $\mathcal{O}$ consists of the solid gray polygons as well as the entire region outside $\mathcal{W}$. The regions with diagonal hashes represent the C-space obstacles, formed by the Minkowski sum of $\mathcal{O}$ and the unit-disk robot. Consequently, the free space $\mathcal{F}$ (the white region) is the complement of these C-space obstacles, representing all valid positions for a robot's center. The semi-transparent yellow region illustrates the trace (swept volume) of the robot along its path.

3.1 Multi-Robot Motion Planning (MRMP)

An instance of the MRMP problem is defined by the workspace $\mathcal{W}$ and two sets of points in the free space $\mathcal{F}$: a set of starting positions $S = \{s_1, \ldots, s_m\}$ and a set of target positions $T = \{t_1, \ldots, t_m\}$. The objective is to plan a continuous, collision-free motion for the m robots, each initially positioned at a distinct starting point in S, such that at the end of the motion, every target position in T is occupied by exactly one robot.

Formally, we seek to compute a solution Γ, which is a set of m continuous paths $\gamma_1, \ldots, \gamma_m$, where each path is a function $\gamma_i : [0, 1] \to \mathcal{F}$. These paths must satisfy two main conditions:

1. **Boundary Conditions:** The paths connect the starts to the targets, meaning $\bigcup_{i=1}^{m} \gamma_i(0) = S$ and $\bigcup_{i=1}^{m} \gamma_i(1) = T$.

2. **Collision-Free Motion:** At any point in time $t \in [0, 1]$, no two robots collide. That is, for every $i \neq j$, the distance between $\gamma_i(t)$ and $\gamma_j(t)$ is at least 2.

In the ***labeled*** variant of the problem, the robots are distinguishable, meaning the robot originating at s_i must terminate exactly at t_i. In the ***unlabeled*** variant, which is the primary focus of this thesis, the robots are indistinguishable; any robot can terminate at any target position as long as the sets S and T are fully matched.

The size (or cost) of a solution Γ is denoted by $|\Gamma| = \sum_{\gamma_i \in \Gamma} |\gamma_i|$, where $|\gamma|$ represents the total length of the path γ measured in the L_2 norm. In the optimization version of MRMP, our goal is to find a feasible solution Γ that minimizes the total path length $|\Gamma|$. We denote the minimum possible sum of lengths of the paths in an optimal collision-free motion plan by OPT.

3.2 Separation Assumptions

To make the problem computationally tractable, we impose specific structural assumptions regarding the initial and final configurations of the robots.

- **Obstacles-Separation (ω):** We assume that every start and target position maintains a minimum clearance from the environment. Formally, for every $p \in S \cup T$ and every point $x \in \mathcal{O}$, it holds that $\|x - p\| \geq \omega$.
- **Robots-Separation (ρ):** We assume a minimum pairwise distance between the configurations themselves. Formally, for every pair of distinct points $p_1, p_2 \in S \cup T$, it holds that $\|p_1 - p_2\| \geq \rho$.

3.3 Overlapping, Blocking, and Interrupting Positions

A core conceptual contribution of our weakly-monotone framework relies on precisely quantifying how robots interfere with one another's paths. To do this, we fix an ***overlap parameter*** $0 \leq \varepsilon \leq 1$. Intuitively, ε represents the maximum permitted "penetration depth" between the regions occupied by a static robot and the trace of a moving robot, defining the exact amount of local wiggle room needed.

Let $p, q \in \mathcal{F}$ be two points in the free space. We define the ***overlap*** of p and q to be $\max\{0, 2 - \|p - q\|\}$. If the distance between them is 2 or more (meaning their respective unit disks $D_1(p)$ and $D_1(q)$ do not intersect), their overlap is 0. Otherwise, we say that p and q are ***ε-overlapping*** for $\varepsilon = 2 - \|p - q\|$, which implies $\|p - q\| = 2 - \varepsilon$.

This overlap concept allows us to classify how a static robot at position $p \in \mathcal{F}$ interacts with the continuous path γ of another moving robot (see Figure 4):

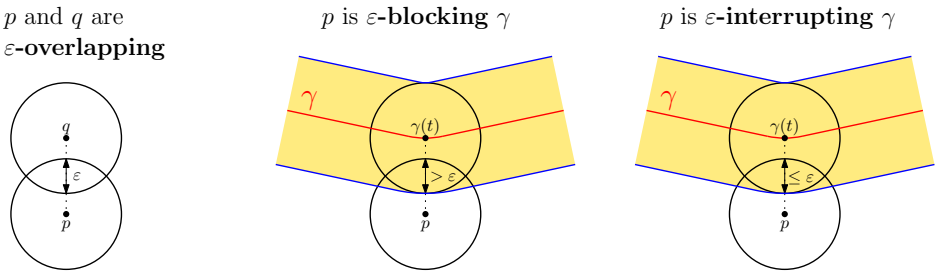


Figure 4: Robot interactions based on the overlap parameter ε. (Left) Two robots with an overlap of ε. (Center) A position p that ε-blocks a path γ. (Right) A position p that ε-interrupts a path γ, indicating it has sufficient wiggle room to be cleared locally.

- **ε-Blocking:** We say that a position $p \in \mathcal{F}$ is ***ε-blocking*** a path γ if there exists some time $t \in [0, 1]$ such that the overlap between p and $\gamma(t)$ is **strictly larger than** ε. Robots occupying blocking positions generally cannot simply step aside; clearing the path usually requires a global "switch" operation.

- **ε-Interrupting:** We say that a point $p \in \mathcal{F}$ is ***ε-interrupting*** a path γ if for every $t \in [0, 1]$, the overlap of p and $\gamma(t)$ is **at most** ε (i.e., it is not ε-blocking). Interrupting robots possess sufficient "wiggle room" within the free space to move slightly away from γ and locally "clear" the path without requiring a switch. Note that if the overlap is 0 for all t, then p is 0-interrupting γ, meaning it does not interfere with the path at all.

Consequently, we define a path γ as ***ε-blocked*** if there exists some start position $s \in S$ that ε-blocks it, and we say that γ is ***ε-interrupted*** if no start position $s \in S$ is ε-blocking it.

For simplicity of the presentation in the subsequent chapters, we sometimes omit ε from the above notions and just say "blocking" or "interrupting". These constraints will dictate the conditions under which interrupting targets exist, blocking robots can be safely switched, and sufficient free space is available for local clearing maneuvers.

3.4 Geodesic Paths and Traces

A geodesic path between two positions $x, y \in \mathcal{F}$ is a shortest length path $\gamma : [0,1] \to \mathcal{F}$ where $\gamma(0) = x$ and $\gamma(1) = y$. Denote by $\mathcal{D}_1(\gamma) = \bigcup_{0 \le t \le 1} \mathcal{D}_1(\gamma(t))$ the ***trace*** of γ (see Figure 5). For $0 \le t_1 \le t_2 \le 1$, denote by $\gamma[t_1, t_2]$ the subpath of γ between $\gamma(t_1)$ and $\gamma(t_2)$.

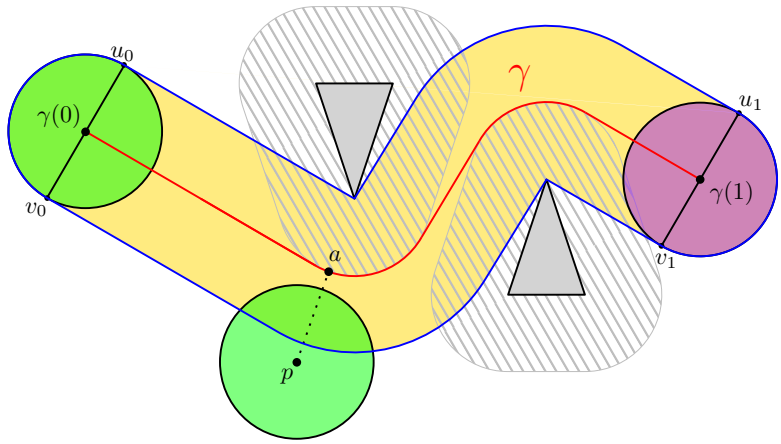


Figure 5: A geodesic path γ in $\mathcal{F}$ from a start position $\gamma(0) \in S$ to a target position $\gamma(1) \in T$ is illustrated in red. The obstacles are in gray, and the trace of γ is highlighted in yellow. A position p is blocking/interrupting γ. The point a is the closest point to p on γ.

4 Constraints for tighter separation bounds

To give an intuition for the required constraints, we first provide a high-level overview of our strategy. The algorithm is recursive, and its input is the overlap parameter ε, a set S of starting positions, a set T of target positions, and a workspace $\mathcal{W}$. In each recursive iteration, one robot is moving to a target position, then S,T and $\mathcal{W}$ are updated accordingly.

The first step is to compute an ***optimal-assignment path set***, as it is defined in [30]:

Definition 1 *Let Γ be a set of m geodesic paths $\gamma_1, \ldots, \gamma_m$, where $\gamma_i : [0,1] \to \mathcal{F}$ for every $1 \leq i \leq m$, and such that $\bigcup_{i=1}^{m} \gamma_i(0) = S$, $\bigcup_{i=1}^{m} \gamma_i(1) = T$. We say that Γ is an optimal-assignment path set for $S, T, \mathcal{W}$ if it minimizes $\sum_{i=1}^{m} |\gamma_i|$ over all such path sets.*

Note that an optimal assignment path set Γ is not necessarily a feasible solution, because it ignores possible collisions between robots. This also means that the sum of paths in Γ is at most the length of an optimal solution to our MRMP problem. Let $\Gamma = \{\gamma_1, \ldots, \gamma_m\}$ be an optimal-assignment path set for $S, T, \mathcal{W}$.

In Section 4.1 we present a useful lemma showing that a blocking robot can enter the geodesic path that it blocks. This lemma will be useful throughout the analysis, and it requires one constraint on the obstacles-separation, and another on the robots-separation.

The next step is to find a target $t \in T$ which does not block any path in Γ (other than the path γ from $s \in S$ to t). Then we can move a robot from s to t along γ, and it will not block other paths in the following iterations. Note that t may still interrupt other paths. In Section 4.2 we show that such a target always exists, when assuming an additional constraint on the robots-separation.

If there is a position $s' \in S$, which blocks the path γ, we need to make a switch, i.e., we construct a new path in $\mathcal{F}$ from s' to t, which is not blocked by any other robot. In Section 4.3 we show how to construct such a switch path γ', under all the constraints defined earlier. If there is no such blocking position s', we simply set $\gamma' \leftarrow \gamma$.

We now have a path γ' from some $s'' \in S$ (which may be either s or s') to the target t, which is not blocked by any robots, but may be interrupted. In Section 4.4 we show how to move a robot from s'' to t along γ', while all interrupting robots (either on start or target positions) move slightly in order to clear the path. For this we introduce two additional constraints, one on the obstacles-separation, and one on the robots-separation.

Finally, we prepare the input for the next recursive iteration of the algorithm, by removing s'' from S, t from T, and $D_{1-\varepsilon}(t)$ from $\mathcal{W}$. Removing $D_{1-\varepsilon}(t)$ from $\mathcal{W}$ ensures that the robot placed on t does not block any path that will be computed in any of the next iterations.

In Chapter 5 we present the complete algorithm based on the building blocks presented in this section, prove its correctness and analyze its running time and approximation factor.

4.1 Geodesics and blocking positions in the free space

In this section we provide some very useful geometric properties of geodesic paths in the free space, which allow us to set the constraints under which a blocking robot can enter a geodesic path via a straight line segment connecting it to the path.

Notice that when $\mathcal{W}$ is a polygonal environment, the boundary of $\mathcal{F}$ consists of straight line segments and arcs of unit circles (see, e.g., [34]). Moreover, if $\mathcal{F}$ contains a point x such that $D_1(x)$ touches a reflex vertex v of $\mathcal{W}$, then the boundary of $\mathcal{F}$ contains an arc of $D_1(v)$.

A geodesic path γ in $\mathcal{F}$ also consists of straight line segments and arcs of unit circles, where the straight line segments are tangent to circular arcs on the boundary of $\mathcal{F}$ that correspond to reflex vertices of $\mathcal{W}$. Thus for any point in the interior of a geodesic path there is a single tangent line. Observe that since γ is a shortest path, if a point $\gamma(t)$ for some $0 \leq t \leq 1$ lies in the interior of $\mathcal{F}$, then it is locally a straight line segment. In other words, we have the following observation:

Observation 2 *Let $\gamma : [0,1] \to \mathcal{F}$ be a geodesic path, and $0 \leq t \leq 1$. If $\gamma[t-\delta, t+\delta]$ is not a straight line segment for any $\delta > 0$, then $\gamma(t)$ is on the boundary of $\mathcal{F}$, and therefore $\mathcal{D}_1(\gamma(t))$ touches the boundary of $\mathcal{W}$ at a point x. Moreover, $\overline{\gamma(t)x}$ is orthogonal to the tangent of γ at the point $\gamma(t)$.*

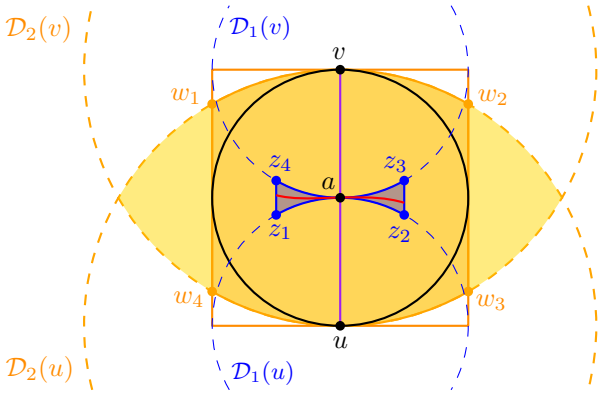


Figure 6: The region $L(a, \overline{uv})$ is in dark yellow. The funnels $F_1(a, \overline{uv}), F_2(a, \overline{uv})$ are in blue.

The cut-out lens and the funnel ports. Consider a point a, and let $\overline{vu}$ be a diametric chord of $\mathcal{D}_1(a)$ (see Figure 6). Let $\mathcal{S}$ be the smallest square enclosing $\mathcal{D}_1(a)$ with two edges parallel to $\overline{vu}$ (its edges are of length 2). Let $L(a, \overline{uv}) := \mathcal{D}_2(v) \cap \mathcal{D}_2(u) \cap \mathcal{S}$. Denote by w_1, w_2, w_3, w_4 the vertices of $L(a, \overline{uv})$ such that w_1, w_2 are on the boundary of $D_2(u)$ and w_3, w_4 are on the boundary of $D_2(v)$. Let z_1 (resp. z_2) be the intersection point of $D_1(u)$ with the segment $\overline{w_1 u}$ (resp. $\overline{w_2 u}$). Similarly, let z_3 (resp. z_4) be the intersection point of $D_1(v)$ with the segment $\overline{w_3 v}$ (resp. $\overline{w_4 v}$). Denote by $F_1(a, \overline{uv})$ (resp. $F_2(a, \overline{uv})$) the funnel defined by the arc $\overset{\frown}{az_1}$ of $D_1(u)$ (resp. the arc $\overset{\frown}{az_2}$ of $D_1(u)$), the arc $\overset{\frown}{az_4}$ of $D_1(v)$ (resp. the arc $\overset{\frown}{az_3}$ of $D_1(v)$), and the segment $\overline{z_1 z_4}$ (resp. $\overline{z_2 z_3}$).

Intuitively, the claim below states that the trace of any path that is contained in the hourglass $F_1(a, \overline{uv}) \cup F_2(a, \overline{uv})$ and crosses $\overline{z_1 z_4}$ and $\overline{z_3 z_2}$ would "cover" the entire region $L(a, \overline{uv})$. It follows from the fact that when rolling a unit disk on the boundary of $L(a, \overline{uv})$, we always cover a segment that connects the upper and lower arcs of the hourglass $F_1(a, \overline{uv}) \cup F_2(a, \overline{uv})$.

Claim 3 *Let π be a continuous path in $F_1(a, \overline{uv}) \cup F_2(a, \overline{uv})$ that starts at a point on $\overline{z_1 z_4}$ and ends at a point on $\overline{z_3 z_2}$. Then $L(a, \overline{uv}) \subseteq D_1(\pi)$.*

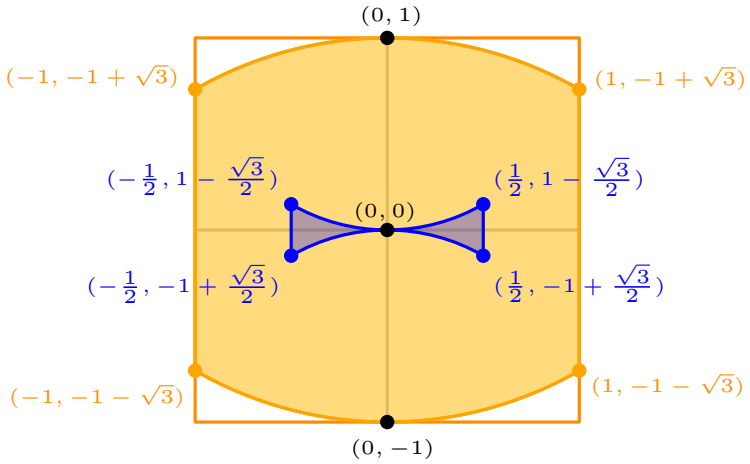


Figure 7: The coordinates assuming that $a = (0,0)$.

Proof. We show that for any point in $L(a, \overline{uv})$ there exists a point on π within distance 1 from it. First, notice that since $F_1(a, \overline{uv}) \cap F_2(a, \overline{uv}) = \{a\}$, the path π must include a, and thus the claim is true for any point $p \in D_1(a)$. We therefore need to prove the claim for points in $L(a, \overline{uv}) \setminus D_1(a)$.

Assume w.l.o.g. that $a = (0,0)$, $v = (1,0)$, $u = (-1,0)$ (see Figure 7). Then the vertices of $L = L(a, \overline{uv})$ are then $w_1 =, w_2 = (1, -1 + \sqrt{3}), w_3 = (1, -1 - \sqrt{3}), w_4 = (-1, -1 - \sqrt{3})$. One can also verify that $z_1 = (-\frac{1}{2}, -1 + \frac{\sqrt{3}}{2}), z_2 = (\frac{1}{2}, -1 + \frac{\sqrt{3}}{2}), z_3 = (\frac{1}{2}, 1 - \frac{\sqrt{3}}{2}), z_4 = (-\frac{1}{2}, 1 - \frac{\sqrt{3}}{2})$.

Because w_1 is on the boundary of $D_2(u)$ we have $\|w_1 - u\| = 2$, and because z_1 is on the boundary of $D_1(u)$ we have $\|z_1 - u\| = 1$. Therefore $\|w_1 - z_1\| = 1$, so $w_1 \in D_1(z_1)$. By symmetric arguments, we have $w_i \in D_1(z_i)$ for every $1 \leq i \leq 4$. Moreover, one can verify that $w_1 \in D_1(z_4)$, $w_2 \in D_1(z_3)$, $w_3 \in D_1(z_2)$ and $w_4 \in D_1(z_1)$.

Consider the quadrilateral $Q_{1,4} = w_1 z_4 z_1 w_4$, and notice that $Q_{1,4} \subseteq D_1(z_1) \cap D_1(z_4)$. Thus for any point $p \in Q_{1,4}$ we have $\overline{z_1 z_4} \in D_1(p)$. Similarly, every point p in the quadrilateral $Q_{2,3} = w_2 z_3 z_2 w_3$ we have $\overline{z_2 z_3} \in D_1(p)$. We conclude that every point in $Q_{1,4} \cup Q_{2,3}$ contains a point in π (either the starting or the ending point).

Finally, consider the region T_1 enclosed by the segments $\overline{w_1 z_1}, \overline{v z_1}$ and the arc $\overset{\frown}{w_1 v}$. Notice that $D_1(z_4)$ contains T_1. Now, because $T_1 \subseteq D_2(u)$, for every point $p \in T_1$ we have $D_1(p) \cap D_1(u) \neq \emptyset$. Moreover, since T_1 is between the radii $\overline{w_1 u}$ and $\overline{vu}$ of $D_2(u)$, the intersection $D_1(p) \cap D_1(u)$ must contain a point on the arc $\overset{\frown}{z_1 a}$ of $D_1(u)$. We conclude that for every

point in T_1 there is a segment $\overline{z_4p'}$ such that p' is on $\overset{\frown}{z_1a}$, and $\overline{z_4p'} \subseteq D_1(p)$. Since π is a continuous path in $F_1(a) \cup F_2(a)$ from a point on $\overline{z_1z_4}$ to a point on $\overline{z_3z_2}$, it must cross the segment $\overline{z_4p'}$, and therefore there is a point on π within distance 1 from p. By symmetric arguments, the claim is also true for the regions T_2, T_3, T_4 that correspond to w_2, w_3, w_4, respectively.

The claim now follows because $L(a, \overline{uv}) = D_1(a) \cup Q_{1,4} \cup Q_{2,3} \cup \bigcup_{i=1}^{4} T_i$. □

Next, we use Claim 3 to show that given a point a on a geodesic path γ, which is far enough from being an endpoint of γ, there is always a region $L(a, \overline{uv})$ that is contained in the trace of γ.

Claim 4 *Let $\gamma : [0,1] \to \mathcal{F}$ be a geodesic path, and let $a = \gamma(t)$ for some $0 \leq t \leq 1$. Denote by $\overline{vu}$ the diametric chord of $\mathcal{D}_1(a)$ perpendicular to the tangent ℓ of γ through a. If $\gamma(0), \gamma(1) \notin F_1(a, \overline{uv}) \cup F_2(a, \overline{uv})$, then $L(a, \overline{uv}) \subseteq \mathcal{D}_1(\gamma)$.*

Proof. We show that if $\gamma(0), \gamma(1) \notin F_1(a, \overline{uv}) \cup F_2(a, \overline{uv})$, then γ has a continuous subpath π that starts at a point on $\overline{z_1z_4}$ and ends at a point on $\overline{z_2z_3}$, and π is completely contained in $F_1(a, \overline{uv}) \cup F_2(a, \overline{uv})$. Then by applying Claim 3 we get that $L(a, \overline{uv}) \subseteq D_1(\pi) \subseteq D_1(\gamma)$.

Assume w.l.o.g., as in the proof of Claim 3, that $a = (0,0)$, $v = (0,1)$, and $u = (0,-1)$. Also assume w.l.o.g. that γ crosses $\overline{uv}$ from left to right.

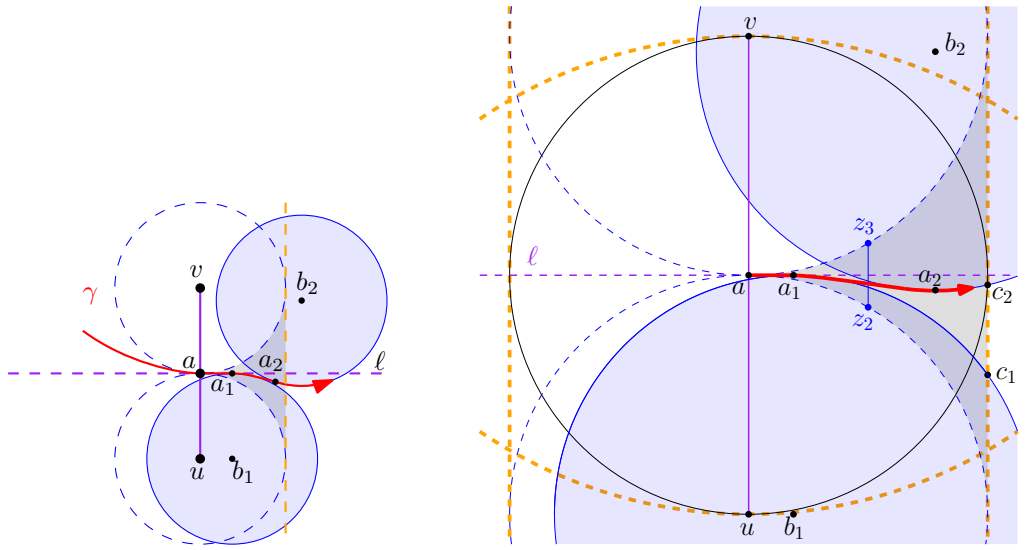


Figure 8: The funnel F_2' is shaded in gray.

Consider the funnel F_2' enclosed by the line $x = 1$ and the disks $D_1(u)$ and $D_1(v)$ (see Figure 8). We will show that $\gamma[t,1]$ does not cross the arcs

of F_2', and because $\gamma(1) \notin F_2(a, \overline{uv})$ this would imply that there exists a value $t \leq t' \leq 1$ such that $\gamma[t, t']$ is contained in $F_2(a, \overline{uv})$ and $\gamma(t') \in \overline{z_2 z_3}$. The other part of the subpath of γ in $F_1(a, \overline{uv})$ then follows by symmetric arguments.

Let $t_1 \geq t$ be the largest value for which $\gamma[t, t_1]$ is a straight line segment. Because $\overline{uv}$ is perpendicular to ℓ, and ℓ is also tangent to both $D_1(u)$ and $D_1(v)$, we get that $\gamma[t, t_1] \subseteq \ell$. If $t_1 = 1$, then clearly $\gamma[t, 1]$ crosses $\overline{z_2 z_3}$ and we are done. Otherwise, by Observation 2 the point $a_1 = \gamma(t_1)$ must lie on the boundary of $\mathcal{F}$, and $\mathcal{D}_1(a_1)$ touches the boundary of $\mathcal{W}$ at a point b_1 which lies either exactly below or exactly above a_1. Assume w.l.o.g. that b_1 lies exactly below ℓ. Because b_1 is on the boundary of $\mathcal{W}$, the path γ does not intersect $\mathcal{D}_1(b_1)$, and the subpath of γ after a_1 has to lie above the arc $\overset{\frown}{a_1 c_1}$, where c_1 is the intersection point of the line $x = 1$ and the boundary of $D_1(b_1)$.

Since γ is a geodesic path, the point a_1 has to be a tangency point of the line through $\overline{aa_1}$ (i.e., ℓ) with the boundary of $\mathcal{F}$. Therefore, there exists some $\varepsilon > 0$ for which $\gamma[t_1, t_1 + \varepsilon]$ is monotonically decreasing. Now let $t_2 \geq t_1$ be the largest value for which $\gamma[t_1, t_2]$ is monotonically decreasing. If $t_2 = 1$, then again we get that $\gamma[t, 1]$ crosses $\overline{z_2 z_3}$ and we are done. Otherwise, by Observation 2 the point $a_2 = \gamma(t_2)$ must lie on the boundary of $\mathcal{F}$, and $\mathcal{D}_1(a_2)$ touches the boundary of $\mathcal{W}$ at a point b_2. Moreover, b_2 has to lie above ℓ, and because b_2 is on the boundary of $\mathcal{W}$, the path γ does not intersect $\mathcal{D}_1(b_2)$. We get that the subpath of γ after a_2 has to lie below the arc $\overset{\frown}{a_2 c_2}$, where c_2 is the intersection point of the line $x = 1$ and the boundary of $D_1(b_2)$.

Notice that $D_1(b_1)$ and $D_1(b_2)$ do not intersect, and therefore c_2 must lie above c_1. Consider the area enclosed by $\overset{\frown}{a_1 c_1}$, $\overline{c_1 c_2}$, $\overset{\frown}{a_2 c_2}$, and $\gamma[t_1, t_2]$. This region contains the subpath of γ from a_1 until it crosses $\overline{c_1 c_2}$. Since this region is also contained in the larger funnel F_2', we get that γ can only cross $F_2(a, \overline{uv})$ at some point on $\overline{z_2 z_3}$. □

The magic radii. Denote $\sigma_\varepsilon = \sqrt{(3 - \sqrt{3} - \varepsilon)^2 + 1}$. This formula describes the obstacles-separation that is sufficient in order for a blocking robot to enter the path that it is blocking.

For two points a, b such that $\mathcal{D}_1(a) \cap \mathcal{D}_1(b) \neq \emptyset$, denote by $\mathcal{R}(a, b)$ the rectangle whose edges are the two diametrical chords of $\mathcal{D}_1(a)$ and $\mathcal{D}_1(b)$ perpendicular to $\overline{ab}$, and the two line segments connecting them on the joint tangents to $\mathcal{D}_1(a)$ and $\mathcal{D}_1(b)$ (See Figure 9).

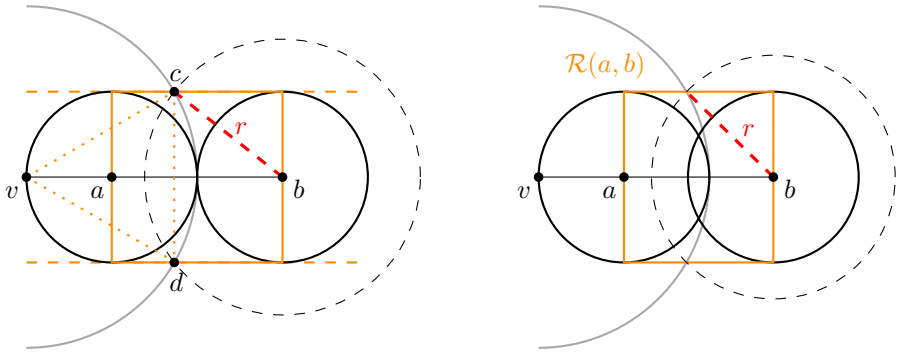


Figure 9: The rectangle $\mathcal{R}(a,b)$ is in orange.

Claim 5 *Let a, b be ε-overlapping, and let v be the intersection point of the line through a and b with the boundary of $\mathcal{D}_1(a)$, which is farther from b. Then $\mathcal{R}(a,b) \subset \mathcal{D}_2(v) \cup \mathcal{D}_{\sigma_\varepsilon}(b)$.*

Proof. Assume w.l.o.g. that v, a, b are on the x-axis, and that $v = (0,0)$, see Figure 9. Then $a = (1,0)$, and $b = (1 + \|b-a\|, 0)$.

Consider the two tangents to $\mathcal{D}_1(a)$ that are parallel to $\overline{ab}$, one above and one below $\overline{ab}$, and denote by c, d the top and bottom intersection points of these tangents with $\mathcal{D}_2(v)$, respectively. Clearly, $D_2(v)$ covers the part of $R(a,b)$ to the left of $\overline{cd}$, and we now show that $\mathcal{D}_r(b)$ covers the part of $R(a,b)$ to the right of $\overline{cd}$.

The distance between the tangents is 2, and therefore $\|c-d\| = 2$. Since $\|v-c\| = \|v-d\| = 2$, the triangle $\triangle vcd$ is an equilateral triangle, and therefore $c = (\sqrt{3}, 1)$ and $d = (\sqrt{3}, -1)$. We have

$$\begin{aligned}\|b-c\| = \|b-d\| &= \sqrt{(1+\|b-a\|-\sqrt{3})^2 + (0-1)^2} \\ &= \sqrt{(1-\sqrt{3}+\|b-a\|)^2 + 1} \\ &= \sqrt{(3-\sqrt{3}-(2-\|b-a\|))^2 + 1}.\end{aligned}$$

The claim follows as $\varepsilon = \max\{0, 2-\|b-a\|\}$. □

Note that if for two points a, b we have $\mathcal{D}_1(a) \cap \mathcal{D}_1(b) \neq \emptyset$, then $\|b-a\| < 2$, and since σ_ε is monotonically decreasing when ε is increasing, by Claim 5 we get that $R(a,b)$ is contained in $\mathcal{D}_2(v) \cap \mathcal{D}_{\sigma_0}(b)$ (where $\sigma_0 = \sqrt{13-6\sqrt{3}}$). This is where the bound conjectured by Solovey et al. [30] is coming from, however, note that it requires the blocking robot to enter the

path through a locally closest point (and not as defined in [30]), as we will clarify later.

We are now ready to set the constraints which allow a blocking robot to enter the path that it is blocking. We have one constraint on the obstacles-separation, and another on the robots-separation.

Constraint 1 $\omega \geq \sqrt{(3-\sqrt{3}-\varepsilon)^2+1}$

Constraint 2 $\rho \geq \sqrt{\frac{1}{2}^2+\left(3-\frac{\sqrt{3}}{2}-\varepsilon\right)^2}$

Let $\gamma : [0,1] \to \mathcal{F}$ be a geodesic path. We say that a point a on γ is ***locally closest*** to p if there exists $\delta > 0$ such that $D_{\|p-a\|}(p) \cap \gamma[a-\delta, a+\delta] = a$.

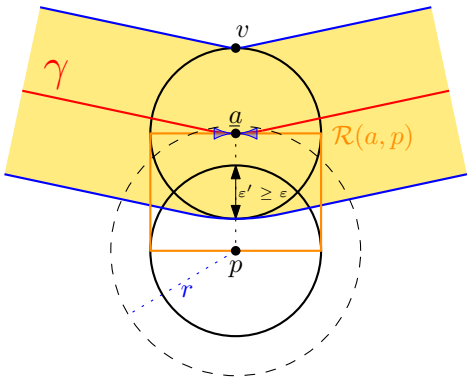


Figure 10: p is ε-blocking γ, a is locally closest to p.

Lemma 6 *Let $\gamma : [0,1] \to \mathcal{F}$ be a geodesic path such that $\gamma(0), \gamma(1) \in S \cup T$, and let $p \in S \cup T$ be a position which is ε-blocking γ. Let $a = \gamma(t)$ be a point on γ which is locally closest to p and such that $\|p-a\| < 2-\varepsilon$. If Constraint 1 and Constraint 2 hold, then $\overline{pa} \subset \mathcal{F}$.*

Proof. First, because $\varepsilon < \varepsilon' = 2 - \|p-a\|$, we have $\sigma_\varepsilon \geq \sigma_{\varepsilon'}$, and therefore $\mathcal{D}_{\sigma_{\varepsilon'}}(p) \subseteq \mathcal{D}_{\sigma_\varepsilon}(p)$. Let v, u be the intersection points of the line through p and a with the boundary of $\mathcal{D}_1(a)$, and assume that v is the point which is farther from p, as in Claim 5 (see Figure 10). Then by Claim 5, $\mathcal{R}(a,p)$ is contained in $\mathcal{D}_2(v) \cup \mathcal{D}_{\sigma_\varepsilon}(p)$. Because $\mathcal{D}_{\sigma_\varepsilon}(p) \subset \mathcal{W}$ (Constraint 1), we have $\mathcal{R}(a,p) \cap \mathcal{D}_{\sigma_\varepsilon}(p) \subset \mathcal{W}$.

Now, since a is a point on γ which is locally closest to p, the segment $\overline{uv}$ is the diametric chord of $\mathcal{D}_1(a)$ perpendicular to the tangent ℓ of γ through a. Notice that by definition, we have $\mathcal{R}(a,p) \cap \mathcal{D}_2(v) \subset L(a,\overline{uv})$. To apply Claim 4 and get that $L(a,\overline{uv}) \subset D_1(\gamma) \subset \mathcal{W}$, we first need to show that $\gamma(0), \gamma(1) \notin F_1(a,\overline{uv}) \cup F_2(a,\overline{uv})$. This follows from Constraint 2: denote $r = \sqrt{\frac{1}{2}^2 + \left(3 - \frac{\sqrt{3}}{2} - \varepsilon\right)^2}$. Because $\gamma(0), \gamma(1) \in S \cup T$, we get that $\|p - \gamma(1)\| \geq r$ and $\|p - \gamma(0)\| \geq r$, and thus $\gamma(0), \gamma(1) \notin S \cup D_r(p)$. One can easily verify that $D_r(p)$ contains the funnels $F_1(a,\overline{uv}) \cup F_2(a,\overline{uv})$ (in fact, this is how we chose r, as the distance between p and e.g. z_3, see Figure 7), and therefore $\gamma(0), \gamma(1) \notin F_1(a,\overline{uv}) \cup F_2(a,\overline{uv})$.

We conclude that $\mathcal{R}(a,p) \cup D_1(a) \cup D_1(p) \subset \mathcal{W}$, and therefore $\overline{pa} \subset \mathcal{F}$. □

Intuitively, Lemma 6 states that a robot blocking γ and positioned on p can move to a along the path $\overline{pa}$, as long as no other robots are standing in its way (see Figure 10). Note that in the following subsections we introduce two more constraints on ρ that together make Constraint 2 redundant, however, Lemma 6 holds already for this weaker constraint and may be of independent interest.

4.2 Existence of an ε-interrupting target

We now define formally a generalization of the standalone goal defined in [30] (Definition 3). Our definition depends on the overlap parameter ε, and it is equivalent to a standalone goal for $\varepsilon = 0$. Let $\Gamma = \{\gamma_1, \ldots, \gamma_m\}$ be an optimal-assignment path set for $S, T, \mathcal{W}$.

Definition 7 *A target position $t \in T$ is an* **ε-interrupting target**, *if for every path $\gamma \in \Gamma$ such that $\gamma(1) \neq t$, it holds that t is not ε-blocking γ.*

To show that an ε-interrupting target always exists, we need an additional constraint.

Constraint 3 $\rho \geq 4 - 2\varepsilon$

The following theorem is a generalization of Theorem 4 in [30]. The proof follows a similar logic, however, we need to take into account some issues that arise from the generalized definition and Constraint 1, Constraint 2, and Constraint 3.

Theorem 8 *Let Γ be an optimal-assignment path set for $S, T, \mathcal{W}$. If Constraint 1, Constraint 2, and Constraint 3 hold, then there exists an ε-interrupting target.*

Proof. Assume by contradiction that there is no ε-interrupting target, that is, for every $t_i \in T$ there exists $\gamma_k \in \Gamma$, $k \neq i$, such that t_i is ε-blocking γ_k. We show that this results in a circular blocking, which leads to a path set of shorter length, in contradiction to the optimality of Γ.

Consider a directed graph that has a vertex v_i for each $t_i \in T$, and for every $1 \leq i, j \leq m$, $i \neq j$, if t_i is ε-blocking γ_j we add the edge (v_i, v_j) to the graph. Because we assume that there is no ε-interrupting target, every vertex has in-degree at least 1, and thus the graph must contain a cycle. We denote this cycle by $(v_1, v_2, \ldots, v_\ell)$, and assume w.l.o.g. that for every $1 \leq i \leq \ell$, the path γ_i is from s_i to t_i, i.e., $\gamma_i(0) = s_i$ and $\gamma_i(1) = t_i$.

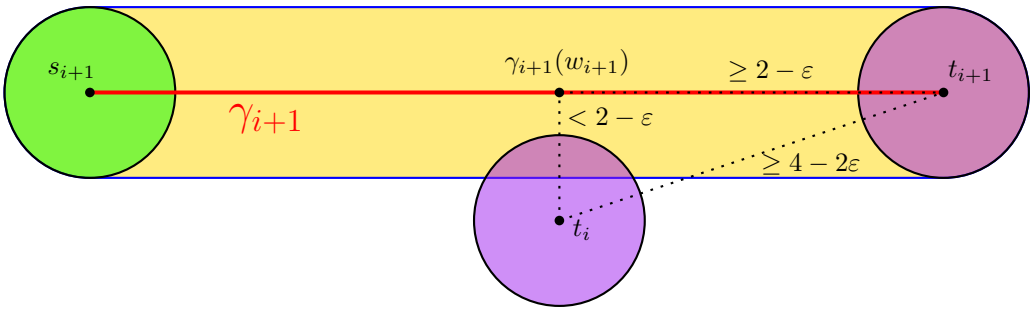


Figure 11: The path γ'_{i+1} from s_{i+1} to t_i is shorter than γ_{i+1}.

Consider an edge (v_i, v_{i+1}) of the cycle, which corresponds to γ_{i+1} being ε-blocked by t_i. Let $\gamma_{i+1}(w_{i+1})$ for $w_{i+1} \in [0,1]$ be the point on γ_{i+1} closest to t_i, then the overlap of $\gamma_{i+1}(w_{i+1})$ and t_i is strictly larger than ε, and thus $\|t_i - \gamma_{i+1}(w_{i+1})\| < 2 - \varepsilon$. Let γ'_{i+1} be the path from s_{i+1} to t_i that follows γ_{i+1} from s_{i+1} until reaching $\gamma_{i+1}(w_{i+1})$, and then continue on the straight line segment from $\gamma_{i+1}(w_{i+1})$ to t_i (see Figure 11). Because Constraint 1 and Constraint 2 hold, we can apply Lemma 6 on γ_{i+1}, t_i and $\gamma_{i+1}(w_{i+1})$, and get that $\overline{\gamma_{i+1}(w_{i+1})\, t_i} \subseteq \mathcal{F}$, and therefore $\gamma'_{i+1} \subseteq \mathcal{F}$. By Constraint 3, $\|t_i - t_{i+1}\| \geq 4 - 2\varepsilon$, and thus by the triangle inequality we have

$$\|\gamma_{i+1}(w_{i+1}) - t_{i+1}\| \geq \|t_i - t_{i+1}\| - \|t_i - \gamma_{i+1}(w_{i+1})\| > 4 - 2\varepsilon - (2 - \varepsilon) = 2 - \varepsilon.$$

We thus have $\|\gamma_{i+1}(w_{i+1}) - t_{i+1}\| > 2 - \varepsilon > \|\gamma_{i+1}(w_{i+1}) - t_i\|$, and since the prefixes of γ_{i+1} and γ'_{i+1} are identical, we get $|\gamma'_{i+1}| < |\gamma_{i+1}|$.

We conclude that for every path γ_i, $1 \leq i \leq \ell$, that corresponds to an edge of the cycle, we can construct a path $\gamma'_i \subset \mathcal{F}$ from s_{i+1} to t_i (or from s_1 to t_ℓ

when $i = \ell$) such that $|\gamma_i'| < |\gamma_i|$. This results in a valid path set of shorter total length, in contradiction to the optimality of Γ. □

4.3 Dealing with a blocked geodesic path

Let $\Gamma = \{\gamma_1, \ldots, \gamma_m\}$ be an optimal-assignment path set for $S, T, \mathcal{W}$, and assume w.l.o.g. that $\gamma_i(0) = s_i$ for every $1 \leq i \leq m$. Consider a geodesic path γ_i from s_i to t_j, and assume that it is ε-blocked. Our goal in this section is to describe the switching process: we find another start point $s_k \in S$ and a path $\gamma_k' \subset \mathcal{F}$ from s_k to t_j such that γ_k' is not ε-blocked. Moreover, we describe a path in $\mathcal{F}$ from s_i to t_ℓ, where $t_\ell = \gamma_k(1)$ (see Figure 12). The sum of lengths of these new paths will not be much longer than the original.

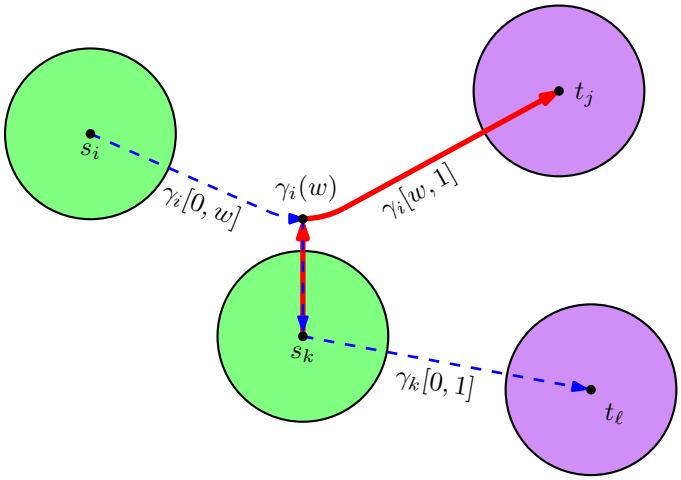


Figure 12: The start position s_k is the last to ε-block γ_i. The switch paths are γ_i' (dashed blue) and γ_k' (red).

For a position $s_k \in S$ that ε-blocks γ_i, denote by $0 \leq w_{i,k} \leq 1$ the largest value such that $\gamma_i(w_{i,k})$ is the point on γ_i locally closest to s_k. We say that $s_k \in S$ is the ***last to ε-block*** γ_i if for any $s_x \in S \setminus \{s_k\}$ that ε-blocks γ_i, we have $w_{i,k} \geq w_{i,x}$ (i.e. the last locally closest point to s_k on γ_i is the closest to $\gamma_i(1)$ along γ_i).

Let s_k be the last to ε-block γ_i, and set $w = w_{i,k}$. Consider the paths $\gamma_k' = \overline{s_k\gamma_i(w)} \circ \gamma_i[w, 1]$ and $\gamma_i' = \gamma_i[0, w] \circ \overline{\gamma_i(w)s_k} \circ \gamma_k$ (where γ_k is the geodesic path from s_k to t_ℓ, see Figure 12). We call γ_i' and γ_k' the ***switch paths*** of i and k, respectively. Notice that $\|s_k - \gamma_i(w)\| < 2 - \varepsilon$, and therefore $|\gamma_k'| +$

$|\gamma_i'| < |\gamma_k| + |\gamma_i| + 4 - 2\varepsilon$. The following theorem will help us in showing that the switch paths are not blocked.

Theorem 9 *Assume that Constraint 1, Constraint 2, and Constraint 3 hold. Then (i) both γ_i' and γ_k' are in $\mathcal{F}$, (ii) $\overline{s_k\gamma_i(w)}$ is not ε-blocked by any $p \in T \cup S \setminus \{s_k\}$, (iii) γ_k' is not ε-blocked by any $s_j \in S \setminus \{s_k\}$, and (iv) if t_j is not ε-blocking γ_k then it does not ε-block γ_i'.*

Proof. To show that (i) holds we only need Constraint 1 and Constraint 2 in order to apply Lemma 6, then we have $\overline{s_k\gamma_i(w)} \subset \mathcal{F}$, and therefore both γ_i' and γ_k' are also contained in $\mathcal{F}$.

For (ii) we also need Constraint 3. Assume by contradiction that there exists $p \in T \cup S \setminus \{s_k\}$ that ε-blocks $\overline{s_k\gamma_i(w)}$. Then there exists a point x on $\overline{s_k\gamma_i(w)}$ such that $\|x - p\| < 2 - \varepsilon$. Because $\|s_k - x\| \leq \|s_k - \gamma_i(w)\| < 2 - \varepsilon$, we get by the triangle inequality that $\|s_k - p\| \leq \|s_k - x\| + \|x - p\| < 4 - 2\varepsilon$, in contradiction to Constraint 3.

To show that (iii) holds, we only need to prove that $\gamma_i[w, 1]$ is not ε-blocked by any position $s_j \in S \setminus \{s_k\}$. Here we also use Constraint 3. Assume by contradiction that there exists $s_x \in S \setminus \{s_k\}$ that ε-blocks $\gamma_i[w, 1]$. Then there exists a value $w \leq w' \leq 1$ such that s_x and $\gamma(w')$ are ε'-overlapping, for some $\varepsilon' > \varepsilon$. Because s_k is the last to ε-block γ_i, we have $w_{i,x} \leq w$. Since $w_{i,x}$ is the largest value such that $\gamma_i(w_{i,x})$ is the point on γ_i locally closest to s_x, it must be that s_x and $\gamma_i(w)$ are ε''-overlapping for some $\varepsilon'' > \varepsilon$. Therefore by the triangle inequality, $\|s_k - s_x\| \leq \|s_k - \gamma_i(w)\| + \|\gamma_i(w) - s_x\| < 2 - \varepsilon + 2 - \varepsilon = 4 - 2\varepsilon$, in contradiction to Constraint 3.

For (iv), we only need to show that t_j does not ε-block $\gamma_i[0, w]$. We again use Constraint 3. Assume by contradiction that t_j blocks $\gamma_i[0, w]$. Then there exists a value $0 \leq w' \leq w$ such that t_j and $\gamma(w')$ are ε'-overlapping, for some $\varepsilon' > \varepsilon$. If $\gamma(w')$ is not locally closest to t_j, then since by (ii) t_j and $\gamma(w)$ do not ε-overlap, there must be a point $0 \leq w'' < w$ locally closest to t_j such that $\|t_j - \gamma(w'')\| < 2 - \varepsilon$. We can therefore apply Lemma 6 and get that $\overline{t_j\gamma(w'')} \subset \mathcal{F}$, in contradiction to γ being a shortest path between s_i and t_j. □

4.4 Dealing with robots interrupting a path

In this subsection, we discuss how to move a robot along an ε-interrupted path. We describe simple motion paths for the interrupting robots to clear

the way, and we analyze the extra distance traveled by the interrupting robots. We note that shorter paths can be shown for the interrupting robots, however, as this does not affect the correctness of the algorithm, we chose to keep this part simple and describe a simple (yet rather short) motion path.

As mentioned at the beginning of this section, our motion plan consists of a weakly-monotone collision free path set, that is, robots move to targets according some ordering one by one; when one robot travels from its starting position to some target, the other robots must stay in a small disk centered at their current position (which is either a start or a target position). To ensure that an interrupting robot is able to clear the way while remaining in a disk of radius $1+\varepsilon$ around a start/target position, we introduce two additional constraints.

Constraint 4 $\omega \geq 1+\varepsilon$

Constraint 5 $\rho \geq 2+\varepsilon$

Consider a robot A traveling at constant speed along a path γ_A from start position $s \in S$ to some target position $t \in T$. Let $p \in S \cup T$ be either a start or a target position, occupied by another robot B, which is ε-interrupting γ_A. We describe a motion plan for B by constructing a ***clearance path*** $\tilde{\gamma}_B : [0,1] \to \mathbb{R}^2$, along which the robot B will travel with constant speed, as follows.

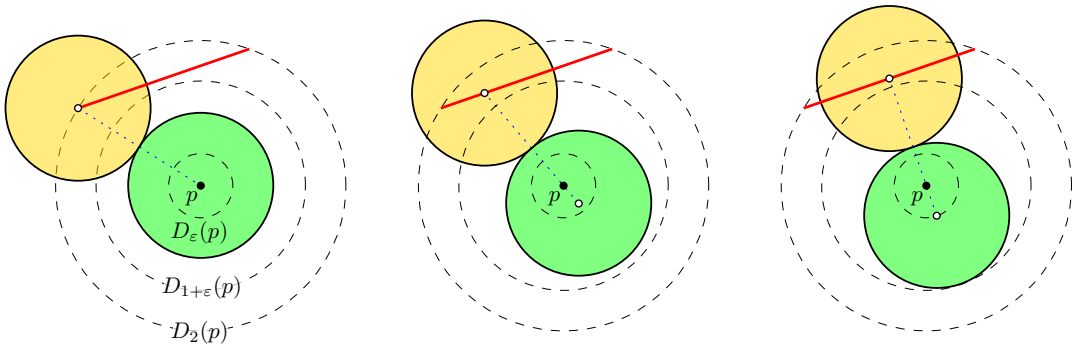


Figure 13: Left: the robot B is initially positioned at p. In red is a maximal subpath of γ_A where B needs to clear the way. Middle, right: the robot B moves inside $\mathcal{D}_\varepsilon(p)$ while maintaining distance exactly 2 from the robot A.

Let $\gamma_A[w_1, w_2]$ be a maximal subpath where $\|\gamma_A(w) - p\| \leq 2$ for every $w \in [w_1, w_2]$, that is, when A travels along $\gamma[w_1, w_2]$, the robot B must clear

the way. Note that there could be more than one such maximal subpath for B. We define the corresponding subpath $\tilde{\gamma}_B : [w_1, w_2] \to \mathbb{R}^2$ for B using a polar coordinate system, where p is the pole. Denote by r_x the radius and by θ_x the angle of a point x, and let $\tilde{\gamma}_B(w) := \left(2 - r_{\gamma_A(w)}, \theta_{\gamma_A(w)} + \pi\right)$.

The next lemma shows that the clearance path of B is feasible and relatively short.

Lemma 10 *$\tilde{\gamma}_B(w)[w_1, w_2]$ is a continuous path that starts and ends at p, such that (i) $\tilde{\gamma}_B[w_1, w_2] \subset D_\varepsilon(p) \subset \mathcal{F}$, and (ii) $|\tilde{\gamma}_B[w_1, w_2]| \leq |\gamma_A[w_1, w_2]|$ for $0 \leq \varepsilon \leq 1$.*

Proof. Observe that indeed $\tilde{\gamma}_B(w)[w_1, w_2]$ is a continuous path because the function $f(w) = (r_{\gamma_A(w)}, \theta_{\gamma_A(w)})$ for $w \in [w_1, w_2]$ is continuous. Notice that for all $w \in [w_1, w_2]$, we have $\|\gamma_A(w) - \tilde{\gamma}_B(w)\| = 2$, and therefore it starts at p when the path γ_A enters the disk $D_2(p)$, and ends at p when the path γ_A leaves the disk $D_2(p)$.

Because p is ε-interrupting γ_A, the largest distance that it has to move away from p to keep $\|\gamma_A(w) - \tilde{\gamma}_B(w)\| = 2$ is ε, and therefore we have $\tilde{\gamma}_B[w_1, w_2] \subset D_\varepsilon(p)$. By Constraint 4, $D_{1+\varepsilon}(p) \subset \mathcal{W}$ and therefore $D_\varepsilon(p) \subset \mathcal{F}$.

Notice that throughout the motion we keep the angle between $\gamma_A(w)$ and $\tilde{\gamma}_B(w)$ (relative to the pole p) exactly π. Thus, the angle of the paths changes at the same pace as w changes. On the other hand, the sum of radii of $\gamma_A(w)$ and $\tilde{\gamma}_B(w)$ throughout the motion is exactly 2, but since $\varepsilon \leq 1$ we get that the radius of $\tilde{\gamma}_B(w)$ is at most the radius of $\gamma_A(w)$ for all $w \in [w_1, w_2]$. Therefore $|\tilde{\gamma}_B[w_1, w_2]| \leq |\gamma_A[w_1, w_2]|$. □

Recall that B may interrupt γ_A in more than one maximal subpath, so overall we define the clearance path $\tilde{\gamma}_B : [0, 1] \to \mathbb{R}^2$ as

$$\tilde{\gamma}_B(w) = \begin{cases} p, & \text{if } \|p - \gamma_A(w)\| \geq 2, \\ \left(2 - r_{\gamma_A(w)}, \theta_{\gamma_A(w)} + \pi\right), & \text{if } \|p - \gamma_A(w)\| < 2. \end{cases}$$

In addition, we construct such a clearance path for any other robot that interrupts γ_A. The following theorem shows that clearance paths are collision-free, assuming Constraint 1, Constraint 2, Constraint 3, Constraint 4, and Constraint 5.

Theorem 11 *Let A be a robot traveling at constant speed along a path γ_A from start position $s \in S$ to some target position $t \in T$, and let $\{\tilde{\gamma}_B \mid B \neq A\}$ be the*

set of clearance paths defined for the other robots. If Constraint 1, Constraint 2, Constraint 3, Constraint 4, and Constraint 5 hold for $0 \leq \varepsilon \leq 1$*, then the motion plan that corresponds to the paths* $\gamma_A \cup \{\tilde{\gamma}_B \mid B \neq A\}$ *is collision-free.*

Proof. Let $B, C \neq A$ be two robots placed on $p_B, p_C \in S \cup T$, respectively. We show that the motion plan that corresponds to the paths $\tilde{\gamma}_B, \tilde{\gamma}_C$ and γ_A is collision-free.

By the construction, for every robot $B \neq A$ the motion plan that corresponds to the paths γ_A and $\tilde{\gamma}_B$ is collision-free. We now need to prove that two robots $B, C \neq A$ traveling along their respective paths $\tilde{\gamma}_B$ and $\tilde{\gamma}_C$ with constant speed are not colliding.

Recall that by the construction of the paths we have $\tilde{\gamma}_B \subset D_\varepsilon(p_B)$ and $\tilde{\gamma}_C \subset D_\varepsilon(p_C)$. By Constraint 5, $|p_B - p_C| \geq 2 + \varepsilon$. Therefore, for every $0 \leq w \leq 1$ such that either $\tilde{\gamma}_B(w) = p_B$ or $\tilde{\gamma}_C(w) = p_C$, the robots do not collide. Consider a value $0 \leq w \leq 1$ for which both $\tilde{\gamma}_B(w) \neq p_B$ and $\tilde{\gamma}_C(w) \neq p_C$. Since B moves along the ray from γ_A to p_B, and C moves along the ray from γ_A to p_C, the distance between A and B can only grow larger at each step of time, and thus B and C do not collide. □

5 Unlabeled MRMP with Improved Separation Trade-offs

In this section we present and analyze the algorithm based on the tools developed in the previous section. The input for the algorithm is an overlap parameter ε, a workspace $\mathcal{W}$, a set S of starting positions and a set T of target positions, such that Constraint 1, Constraint 2, Constraint 3, Constraint 4, and Constraint 5 hold. In addition, we assume that each connected component of $\mathcal{F}$ contains an equal number of start and target positions (as otherwise, there is no solution).

For the simplicity of presentation and analysis, in this section we provide an iterative version of the algorithm. In the i'th iteration, for $1 \leq i \leq m$, the algorithm constructs a collision-free motion plan for all the robots over the time interval $[i-1, i]$, in which a single robot is moving from a start position to a target position along an ε-interrupted path (Section 4.3), and the other robots may move in their small neighborhood to clear the way (Section 4.4). The output path for a robot A will then be the concatenation $\gamma_A^* : [0, m] \to \mathcal{F}$ of the paths $\gamma_A : [i-1, i] \to \mathcal{F}$ constructed for it in all iterations. We then update S and T according to the path that was chosen, and remove from $\mathcal{W}$ a disk of radius $1 - \varepsilon$ around the target.

At the beginning of this section we gave a high level description of our algorithm, and a more detailed pseudo code is provided in Algorithm 1. Below, we prove that this algorithm returns a collision-free motion plan, with total length $O(\mathrm{OPT})$, where OPT is the optimal size of a solution to the problem.

Theorem 12 *Consider an input $\mathcal{W}$, S, T, and $0 \leq \varepsilon \leq 1$ such that Constraint 1, Constraint 2, Constraint 3, Constraint 4, and Constraint 5 hold. If each connected component of $\mathcal{F}$ contains an equal number of start and target positions, then Algorithm 1 is guaranteed to return a collision-free motion plan Γ^*.*

Input: $\mathcal{W}$, S, T, and $0 \leq \varepsilon \leq 1$ such that Constraint 1, Constraint 2, Constraint 3, Constraint 4, and Constraint 5 hold
Output: A collision-free motion plan

- For each robot $A \in [m]$, initialize an empty path γ_A^*.
- While $S \neq \emptyset$:
  1. Compute an optimal-assignment path set Γ for $S, T, \mathcal{W}$.
  2. Find an ε-interrupting target t_j, and let $\gamma_i \in \Gamma$ be the path from s_i to t_j.
  3. If γ_i is not ε-blocked, set $A \leftarrow i$, $s_A \leftarrow s_i$, $\gamma_A \leftarrow \gamma_i$.
  4. Else, let s_k be the last to ε-block γ_i, and set $A \leftarrow k$, $s_A \leftarrow s_k$, and $\gamma_A \leftarrow \gamma_k'$, where γ_k' is the switch path (Section 4.3).
  5. For all $B \neq A$, set $\gamma_B \leftarrow \tilde{\gamma}_B$, where $\tilde{\gamma}_B$ is the clearance path (Section 4.4).
  6. For each robot $B \in [m]$ set $\gamma_B^* \leftarrow \gamma_B^* \circ \gamma_B$.
  7. Set $\mathcal{W} \leftarrow \mathcal{W} \setminus \mathcal{D}_{1-\varepsilon}(t_j)$, $S \leftarrow S \setminus \{s_A\}$, $T \leftarrow T \setminus \{t_j\}$.
- Return the set of paths $(\gamma_1^*, \ldots, \gamma_m^*)$.

Algorithm 1: Motion Plan for Unlabeled Disks

Proof. First, observe that the solution returned by the algorithm is a path set that matches every start position in S to a unique target in T, because in each iteration we find a path between some $s \in S$ and $t \in T$ and then remove them from S and T for the next iteration.

Consider the first iteration of the algorithm. By Theorem 8, an ε-interrupting target t_j exists, and let γ_i be the path from s_i to t_j. If γ_i is ε-blocked, let $s_k \in S$ be the last to block γ_i, and by Theorem 9 (iii), the switch path γ_k' is in $\mathcal{F}$, and it is not blocked by any other robot. Otherwise, γ_i is not blocked. In any case we choose an ε-interrupted path that leads to t_j, and by Theorem 11 the motion plan computed in this iteration is collision-free.

In the following iteration, after removing $D_{1-\varepsilon}(t_j)$ from $\mathcal{W}$, Constraint 1, Constraint 2, Constraint 3, Constraint 4, and Constraint 5 still hold for the updated sets S and T: the robots-separation does not change, so Constraint 2, Constraint 3, and Constraint 5 still hold, and specifically by Constraint 3 for every $p_1, p_2 \in S \cup T$ we have $\|p_1 - p_2\| \geq 4 - 2\varepsilon$. Therefore, for

every $p \in S \cup T \setminus \{t_j\}$ and $x \in D_{1-\varepsilon}(t_j)$ we have $\|p - x\| \geq 3 - \varepsilon$, which satisfies both Constraint 1 ($\|p - x\| \geq \rho_\varepsilon$) and Constraint 4 ($\|p - x\| \geq 1 + \varepsilon$), for any $0 \leq \varepsilon \leq 1$.

However, we also need to take into account the robot that is now positioned on t_j. The main observation that we will use is that t_j does not block any path γ in the free space of $\mathcal{W} \setminus D_{1-\varepsilon}(t_j)$; the reason is that the distance between any point on $a \in \gamma$ and a point $p \in D_{1-\varepsilon}(t_j)$ is at least 1, and therefore the $\|t_j - a\| \geq 2 - \varepsilon$ which means that the overlap between t_j and a is at most ε.

Consider an iteration $a > 1$ of the algorithm, and denote by $\mathcal{W}_a, S_a, T_a$ the sets $\mathcal{W}, S, T$ at the beginning of iteration a. The first step is to compute an optimal-assignment path set Γ_a. Let Γ_{a-1} be the optimal-assignment path set for $\mathcal{W}_{a-1}, S_{a-1}, T_{a-1}$. At the beginning of iteration a, we have $\mathcal{W}_a = \mathcal{W}_{a-1} \setminus \mathcal{D}_{1-\varepsilon}(t_j)$ and $T_a = T_{a-1} \setminus \{t_j\}$, where t_j is an ε-interrupting target in Γ_{a-1}. Since t_j does not block any path in Γ_{a-1}, the disk $\mathcal{D}_{1-\varepsilon}(t_j)$ does not overlap with the trace of any path in $\Gamma_{a-1} \setminus \{\gamma_i\}$. Therefore, any path $\gamma \in \Gamma_a \setminus \{\gamma_i\}$ is in the free space of $\mathcal{W}_a$. If γ_i was not blocked in iteration $a - 1$, then $S_a = S_{a-1} \setminus \{s_i\}$, and thus $\Gamma_a \setminus \{\gamma_i\}$ corresponds to a matching of S_a and T_a, which implies that Γ_a exists. Otherwise, $S_a = S_{a-1} \setminus \{s_k\}$, where $s_k \in S_{a-1}$ is the last to ε-block γ_i. Let t_ℓ be the target that was matched to it in Γ_{a-1}, and let $\gamma_k \in \Gamma_{a-1}$ be the path from s_k to t_ℓ. Observe that $\Gamma_{a-1} \setminus \{\gamma_i, \gamma_k\}$ corresponds to a matching between $S_a \setminus \{s_i, s_k\}$ and $T_{a-1} \setminus \{t_k, t_\ell\}$, and thus to have a matching between S_a and T_a we need to find a path in the free space of $\mathcal{W}_a$ between s_i and t_ℓ. We show that the switch path γ_i' from s_i to t_ℓ as defined in Section 4.3, has this property. Recall that γ_i' is a concatenation of $\gamma[1, w]$, $\overline{\gamma_i(w)s_k}$, and γ_k. The paths γ_i, γ_k are in Γ_{a-1} and therefore lie in the free space of $\mathcal{W}_{a-1}$. By Theorem 9(i), the entire path γ_i' is in the free space of $\mathcal{W}_{a-1}$. We thus need to show that the disk $\mathcal{D}_{1-\varepsilon}(t_j)$ does not overlap with the trace of γ_i'. Indeed, since t_j is an ε-interrupting target, it does not block γ_k, and therefore by Theorem 9(iv) it is also not blocking γ_i'.

The next step is to find an ε-interrupting target t_j. For this, notice that Theorem 8 still holds, because it considers only start and target positions in S_a and T_a. Let $\mathcal{A}$ be the set of robots that are already lying on target positions. Notice that we can also apply Theorem 9 on $\mathcal{W}_a, S_a, T_a$ and Γ_a, and we just need to make sure that no robot in $\mathcal{A}$ can block the switch path γ_k'. As mentioned above, this is true in general, that is, for any path $\gamma \in \Gamma_a$ from $s \in S_a$ to $t \in T_a$, no robot in $\mathcal{A}$ can block γ, because the overlap between any such robot and any point on γ is at most ε. In other words,

robots in $\mathcal{A}$ can only interrupt paths in Γ_a.

The final step is using Theorem 11 to construct the clearance paths. Notice that this construction applies for any $p \in S \cup T$, and therefore it can be applied for the ε-interrupted path, even if there were robots in all the start and target positions, except for the endpoints of the path.

Lastly, notice that the concatenation of the paths that correspond to a robot A from all the iterations of the algorithm results in a continuous path. This is because each clearance path starts and ends at the same start/target position. □

Theorem 13 *Algorithm 1 runs in $O(m^4 + m^2n^2)$ time, and returns a solution Γ^* such that $|\Gamma^*| = O(OPT)$, where OPT is the size of an optimal solution.*

Proof. To bound the sum of distances we separate it into two parts - (1) the distances traveled by robots while traveling from their starting position to a target position and (2) the distances of clearance paths.

Starting with (1), let Γ_a be the optimal assignment that the algorithm found in the a'th iteration and let Γ_a^* be the paths assigned in the iterations before the a'th iteration (not including the clearance paths). We claim that $|\Gamma_{a+1} \cup \Gamma_{a+1}^*| \leq |\Gamma_a \cup \Gamma_a^*| + 4 - 2\varepsilon$. To see that, let t_j be the ε-interruption target found in the a'th iteration and let s_i be the starting position such that the path γ_i leads to t_j. If this path is ε-interrupted then $\Gamma_{a+1}^* = \Gamma_a^* \cup \{\gamma_i\}$ and since $\Gamma_a \setminus \{\gamma_i\}$ is a feasible assignment path set in the next iteration then $|\Gamma_{a+1}| \leq |\Gamma_a \setminus \{\gamma_i\}|$ and indeed, $|\Gamma_{a+1} \cup \Gamma_{a+1}^*| \leq |\Gamma_a \cup \Gamma_a^*| \leq |\Gamma_a \cup \Gamma_a^*| + 4 - 2\varepsilon$.

Otherwise γ_i is ε-blocked. Let s_k be the last ε-blocking target. Recall the switch paths defined in Section 4.3. In this case, $\Gamma_{a+1}^* = \Gamma_a^* \cup \{\gamma_k'\}$ and since $(\Gamma_a \setminus \{\gamma_i, \gamma_k\}) \cup \{\gamma_i'\}$ is feasible assignment path set in the next iteration then $|\Gamma_{a+1}| \leq |(\Gamma_a \setminus \{\gamma_i, \gamma_k\}) \cup \{\gamma_i'\}|$ and we get $|\Gamma_{a+1} \cup \Gamma_{a+1}^*| \leq |((\Gamma_a \setminus \{\gamma_i, \gamma_k\}) \cup \{\gamma_i'\}) \cup \Gamma_a^* \cup \{\gamma_k'\}|$. By the definition of the switch paths we get $|\gamma_i| + |\gamma_k| = |\gamma_i'| + 2 - \varepsilon + |\gamma_k'| + 2 - \varepsilon$ and we conclude that $|\Gamma_{a+1} \cup \Gamma_{a+1}^*| \leq |\Gamma_a \cup \Gamma_a^*| + 4 - 2\varepsilon$.

Applying the inequality m times we get that $|\Gamma_m \cup \Gamma_m^*| \leq |\Gamma_1 \cup \Gamma_1^*| + (4 - 2\varepsilon)m$ and since $\Gamma_1^*, \Gamma_m = \emptyset$ and since $|\Gamma_1| \leq$ OPT (as Γ_1 is the best assignment path set even when ignoring collision and OPT is the size of an assignment path set) we conclude that $|\Gamma_m^*| \leq \text{OPT} + (4 - 2\varepsilon)m$.

As for the extra distance traveled by the clearance paths, let γ_A be the path that leads to the ε-interrupting target at some iteration. Let p be a starting/target position occupied by a robot B that interrupts γ_A. For a max-

imal subpath $\gamma_A[w_1, w_2]$ where $\|\gamma_A(w) - p\| \leq 2$ for every $w \in [w_1, w_2]$, the robot on p traveled according to $\tilde{\gamma}_B[w_1, w_2]$. Lemma 10 shows that $\tilde{\gamma}_B[w_1, w_2] \leq \gamma_A[w_1, w_2]$. We mark with c the maximal number of robots that can interrupt γ_A at a single location, that without any separation assumptions its 6, and conclude that $\sum_{B \neq A} |\tilde{\gamma}_B| \leq c|\gamma_A|$. So by adding the extra distance traveled in the clearance paths in all the iterations we get that $|\Gamma^*| \leq (1+c)\Gamma^*_m \leq (1+c)(\text{OPT} + (4-2\varepsilon)m = O(\text{OPT})$. Note that this analysis is not tight; see Section 5.2 for more details.

Regarding the running time, note that except for computing the clearance paths in each iteration, the rest of the operations are changed version of operations of the algorithm in [30] and the implementation analysis in [30] provides a valid analysis of our operations and yields $O\left(m^4 + m^2n^2\right)$ for the changed operations. As for computing the clearance paths, assume in a specific iteration that we have already the path γ_A that leads a robot to an ε-interruption target and we need to compute only the clearance paths. The path consists of segments and arcs and its complexity is bounded by the complexity of $\mathcal{F}$ which is $O\left(m+n\right)$ (as in each iteration we add exactly one disk to $\mathcal{O}$). For each segment or arc computation the clearance path takes $O\left(1\right)$ and totally, for each robot the iteration it takes $O\left(m+n\right)$ time. So, for m robots in m iterations the total running time for computing the clearance paths is $O\left(m^3 + m^2n\right)$ which is absorbed in the $O\left(m^4 + m^2n^2\right)$. □

We conclude with the following theorem.

Theorem 14 *Given a polygonal workspace $\mathcal{W}$ with n edges, a set S of m starting positions and a set T of m target positions, there is an algorithm that computes in $\tilde{O}(m^4 + n^2m^2)$ time a collision-free motion plan of length $O(OPT)$, where OPT is the length of an optimal solution, or reports that such a motion plan does not exist, assuming that $\rho = \max\{4-2\varepsilon, 2+\varepsilon\}$ and $\omega = \max\{1+\varepsilon, \sqrt{(3-\sqrt{3}-\varepsilon)^2+1}\}$.*

5.1 Optimizing the separation bounds

We now optimize the overlap parameter over Constraints 1-5, to achieve either the smallest obstacles-separation or the smallest robots-separation possible. Recall that Constraint 1 and Constraint 4 specify the required obstacles-separation bounds, while Constraint 2, Constraint 3, and Constraint 5 specify the required robots-separation bounds.

The resulting upper envelopes for these bounds, along with their optimal minimum points, are visualized in Figure 14. Specifically, we highlight three interesting values of ε:

- For $\varepsilon = 0$ the motion plan is monotone, and the length of the solution is at most OPT $+ 4m$. In this case, we obtain $\omega = \sqrt{13 - 6\sqrt{3}} \approx 1.614$ and $\rho = 4$.
- For $\varepsilon > 0$, the motion plan is weakly-monotone.
- For $\varepsilon = \frac{15-6\sqrt{3}}{13} \approx 0.354$, the obstacles-separation is minimized, and we obtain $\omega = 1 + \frac{15-6\sqrt{3}}{13} \approx 1.354$ and $\rho = 4 - 2\frac{15-6\sqrt{3}}{13} \approx 3.291$.
- For $\varepsilon = \frac{2}{3}$, the robots-separation is minimized, and we obtain $\omega = 1\frac{2}{3}$ and $\rho = 2\frac{2}{3}$.

5.2 A remark on the length of our solution

Note that in fact, we can bound the size of the solution with the term $(1 + c)(\text{OPT} + (4 - 2\varepsilon)m)$, where c is the maximum number of start/target positions that may intersect a given unit disk under our robots-separation constraints. The term OPT $+ (4 - 2\varepsilon)m$ accounts for the motion of robots from their starting positions to their targets, and the $(1 + c)$ factor comes from the clearance paths.

For example, when the robots-separation is $\rho = 2$ then at most $c = 6$ start/target positions may intersect a given unit disk. When $\varepsilon = 0$ we have $\rho = 4$, so no two start/target positions intersect a single unit disk, and therefore the size of the solution is $OPT + 4m$ — near optimal solution as in [30].

We also note that this bound can be significantly reduced with a tighter analysis for two reasons: (1) even if a robot can collide with c robots at some point, along most of its path it collides with fewer than c; and (2) our analysis of the additional distance traveled in the clearance paths assumes $\varepsilon \leq 1$, but if $\varepsilon \ll 1$, the extra distance traveled during the clearance paths is much smaller.

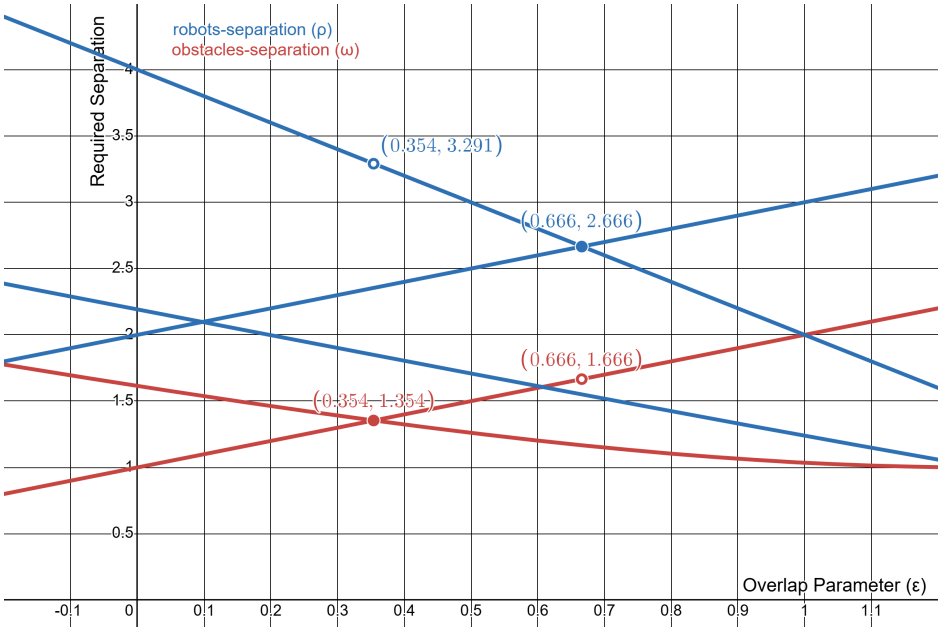


Figure 14: The required separation bounds as a function of the overlap parameter $\varepsilon \in [0, 1]$. The obstacles-separation ω is determined by the upper envelope of Constraints 1 and 4 (red curves). The robots-separation ρ is determined by the upper envelope of Constraints 2, 3, and 5 (blue curves). **Solid points** highlight the minimal possible bound for each parameter: $\omega \approx 1.354$ in the clearance-optimized regime, and $\rho \approx 2.666$ in the density-optimized regime. **Hollow points** indicate the corresponding value of the other parameter at those specific ε values.

6 The Limitations of Monotone and Weakly-Monotone Motion Plans

In this section we present two examples that demonstrate the limitations of (weakly) monotone motion plans. In the first example the obstacles-separation is ≈ 1.614, and we show that no monotone motion plan exists. In the second example the obstacles-separation is ≈ 1.354 and we show that no weakly-monotone motion plan exists. These examples show that in some situations, monotone or even weakly monotone techniques cannot yield a solution when the obstacles-separation is smaller than those presented in this thesis, and therefore a completely different approach is needed for algorithms with lower obstacles-separation assumptions.

6.1 Monotone motion plans

Recall that in a monotone motion plan with $\omega = \sqrt{13 - 6\sqrt{3}} \approx 1.614$, each start/target position must be at least ω away from every obstacle, and the robots move one by one from their starting positions to their target positions, each fully completing its path and never moving again.

Consider Figure 15. The point o is the origin, and the target t_1 is located at $(0, -2 + \delta_1)$ for some arbitrarily small $\delta_1 > 0$. The orange rectangle is $\mathcal{R}(o, t_1)$. The vertex v_1 is located at $(0, 1)$, and the dashed gray disk centered at v_1 has radius 2. The points v_2, v_3 are located at distance exactly 2 from v_1, slightly below the intersection points of $D_2(v_2)$ and $\mathcal{R}(o, t_1)$ (the orange points), whose coordinates are $(\pm 1, 1 - \sqrt{3})$. Note that the distance between these orange points is exactly 2, and that the distance between v_2 and v_3 is strictly smaller than 2. The distance between t_1 and each of the orange points is exactly $\sqrt{1 - (1 - \sqrt{3} + 2 - \delta_1)^2} = \rho_{\delta_1}$, which approaches

$\rho_0 = \sqrt{13 - 6\sqrt{3}}$ when δ_1 approaches 0. We can thus pick $\delta_1 > 0$ and the points v_2, v_3 such that the distance between t_1 and each of v_2, v_3 is $\rho_0 - \delta_2$ for any arbitrarily small $\delta_2 > 0$. The setting for t_2 is symmetric.

Now, observe that s_1, t_1 are in one connected component of $\mathcal{F}$ while s_2, t_2 are in another one (because the distance between v_2 and v_3 is strictly smaller than 2). In a monotone motion plan, one of the robots will move first to a target, and then it will stay there. Assume w.l.o.g. that the robot on s_1 goes first and reaches t_1. To reach t_2, the robot positioned on s_2 must pass through $\overline{v_1 v_2}$ and $\overline{v_1 v_3}$, and therefore the trace of its path must cover the arc of $D_2(v_1)$ between v_2 and v_3. Since $\|t_1 - v_1\| < 2$, the robot positioned on t_1 would always intersect the trace of such path, so s_2 cannot reach t_2.

We conclude that any algorithm that returns a monotone motion plan will not be able to produce an answer to this instance, even though it is easy to see that a feasible solution exists. We note that [2] and [8] present monotone algorithms with $\omega = 1$, but they work only in simple polygons.

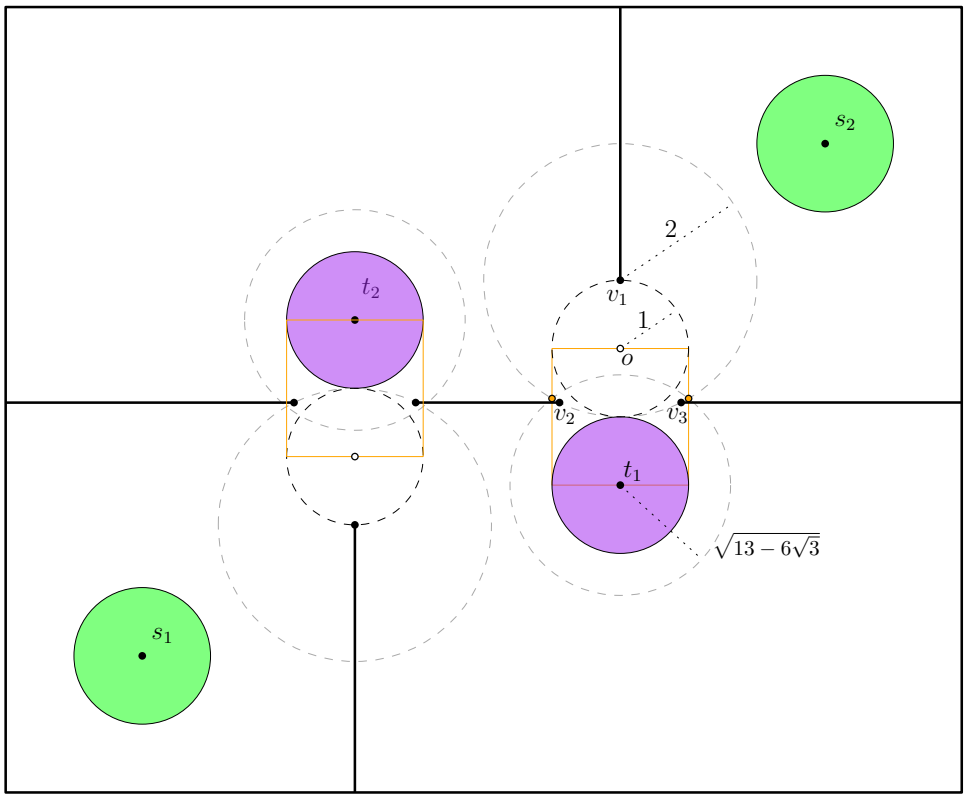


Figure 15: An instance of MRMP with $\omega \approx 1.614$, in which a monotone solution does not exist.

6.2 Weakly-monotone motion plans

Recall that in a weakly-monotone motion plan with respect to radius r, each start/target position must be contained in a disk of radius r, empty of obstacles and other robots. The robots move one by one from their starting positions to their target positions, while other robots must remain inside a disk of radius r centered either at another start or target position.

Let $\varepsilon = \frac{15-6\sqrt{3}}{13} \approx 0.354$ and consider Figure 16. The point o_1 is the origin, and $o_2 = (0, -2)$. The orange rectangle is $\mathcal{R}(o_1, o_2)$. The vertex v_1 is located at $(0, 1)$, and the dashed gray disk centered at v_1 has radius 2. The points v_2, v_3 are located at distance exactly 2 from v_1, slightly below the intersection points of $D_2(v_2)$ and $\mathcal{R}(o_1, o_2)$ (the orange points), whose coordinates are $(\pm 1, 1 - \sqrt{3})$. Note that the distance between these orange points is exactly 2, and that the distance between v_2 and v_3 is strictly smaller than 2.

We place the target t_1 at $(0, -2 + \varepsilon)$, so the distance between t_1 and each of the orange points is exactly $\sqrt{1 - (1 - \sqrt{3} + 2 - \varepsilon)^2} = \rho_\varepsilon$. One can verify that for $\varepsilon = \frac{15-6\sqrt{3}}{13}$ we have $\rho_\varepsilon = 1 + \varepsilon$ (this is how we chose ε in case (ii) of Theorem 14). This means that $\mathcal{D}_{1+\varepsilon}(t_1)$ (marked in dashed gray) contains v_2, v_3, and therefore the largest disk centered at t_1 that does not intersect the boundary of the polygon has radius $1 + \varepsilon - \delta$, for some arbitrarily small $\delta > 0$. Since $\|t_1 - v_1\| = 3 - \varepsilon$, we get that $\mathcal{D}_{1+\varepsilon-\delta}(t_1)$ is contained in $\mathcal{D}_{4-\delta}(v_1)$. The setting for t_2 is symmetric.

Now, as in the previous example, s_1, t_1 are in one connected component of $\mathcal{F}$ while s_2, t_2 are in another one (because the distance between v_2 and v_3 is strictly smaller than 2). In a weakly monotone motion plan, one of the robots will move first to a target, and then it will stay in the disk of radius $1 + \varepsilon - \delta$ around it (this is the largest disk centered at the target and does not intersect the boundary of the polygon). Assume w.l.o.g. that the robot on s_1 goes first. When it reaches t_1, it can still move inside $\mathcal{D}_{1+\varepsilon-\delta}(t_1)$. However, to reach t_2, the robot positioned on s_2 must pass through $\overline{v_1 v_2}$ and $\overline{v_1 v_3}$, and therefore the trace of its path must cover the arc of $D_2(v_1)$ between v_2 and v_3. Since $\mathcal{D}_{1+\varepsilon-\delta}(t_1)$ is contained in $\mathcal{D}_{4-\delta}(v_1)$, the robot positioned on t_1 would always intersect the trace of such path, so s_2 cannot reach t_2.

We conclude that any algorithm that plans monotone motion plan will not be able to produce an answer to this instance although it is easy to see

that a feasible solution exists. Recall that [4] present a weakly-monotone algorithm in which $r = 2$, and the start/target positions do not have to lie at the center of the revolving area. However, we note that it is possible to modify the example in Figure 15 to force the disk $\mathcal{D}_{1+\varepsilon-\delta}(t_1)$ to be the revolving area, even when the target position is not t_1, by blocking it with the boundary of the polygon.

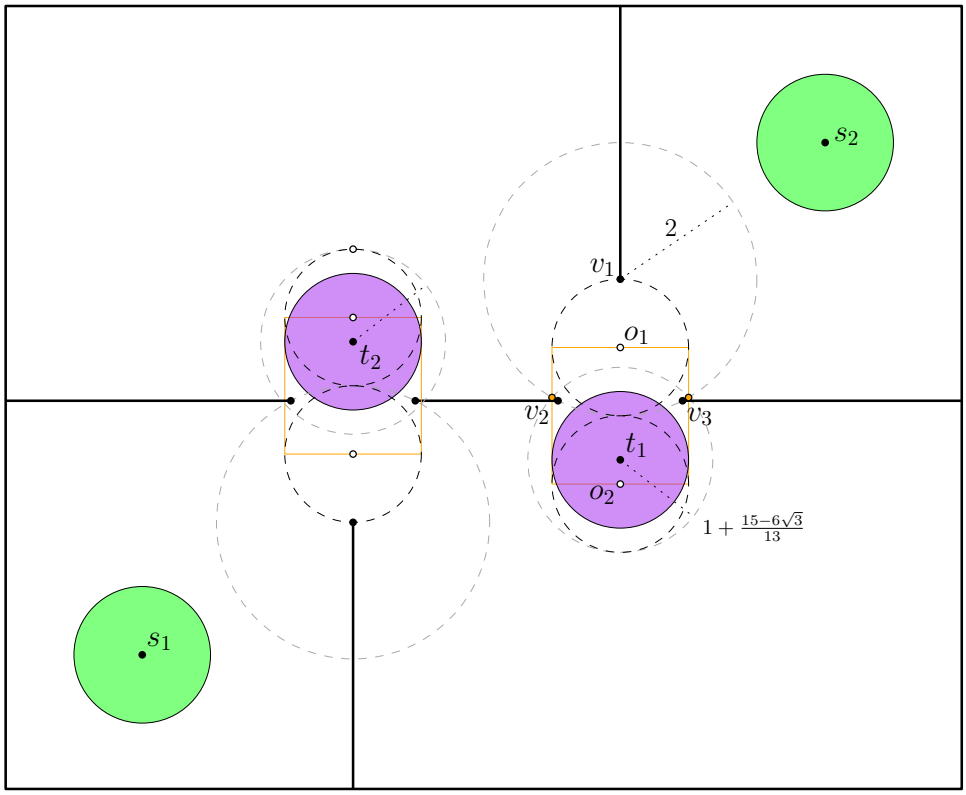


Figure 16: An instance of MRMP with $\omega \approx 1.354$, in which a weakly-monotone solution does not exist.

7 The Exodus Algorithm: Reducing the Robots-Separation Bound

In this chapter we introduce a different algorithmic strategy for MRMP which we call the *Exodus Algorithm*. This strategy is more global in the sense that it also moves robots that do not directly block a path, and therefore it allows for robots-separation of 2, while requiring obstacles-separation of only 3. Whereas the previous algorithm relies on local path-clearing and larger robots-separation, Exodus replaces these local adjustments with a synchronized outward movement of all robots in order to clear the trace of a geodesic path, so that the selected robot can reach its target via this path. Note that in this section, we assume that the workspace $\mathcal{W}$ is a simple polygon. Also note that this strategy works in both the *labeled* and *unlabeled* settings, and ensures collision-free execution under near optimal separation constraints (see Section 7.4). Below, we first provide an overview of the algorithm, then we provide a formal description of the steps of the algorithm with complete proofs.

Algorithm overview. Given the sets $S, T \subset \mathcal{F}$ as before, the first step of the algorithm is to compute an optimal-assignment path set $\Gamma = \{\gamma_1, \ldots, \gamma_m\}$ for $S, T, \mathcal{W}$. Assume w.l.o.g. that $\gamma_i(0) = s_i$ and $\gamma_i(1) = t_i$ for every $1 \leq i \leq m$. The algorithm then proceeds iteratively. In each iteration, we select a target position $t_i \in T$ which is not yet occupied by a robot. We first extend the path γ_i so that its start and end points are on the boundary of $\mathcal{F}$. The extended path partitions $\mathcal{F}$ into maximally connected sub-regions of $\mathcal{F}$, which we call pockets. We then split each pocket using the angle bisectors corresponding to the reflex subchain of γ_i that defines the pocket. We call the cells in this refined partition Exodus cells. Each Exodus cell contains a single straight-line edge of γ_i. All the other robots are then moved outwards from the extended path by two units (which is possible because $\omega = 3$), and the movement direction for each robot is determined

by the Exodus cell in which it lies. This process clears the trace of γ_i, allowing a robot positioned on s_i to move along γ_i until reaching t_i, while all the other robots remain fixed at their offset locations. After the robot reaches its target t_i, all the other robots return to their original positions by applying the inverse movement (which is possible because $\rho = 2$, i.e., no start position is too close to t_i). This iterative "corridor-opening" process motivates the name *Exodus Algorithm*. The procedure repeats until all robots reach their assigned targets. In each iterative step, $m - 1$ robots are moving $4m$ units each, while the selected robot moves along a shortest path to its assigned target (as before, $|\Gamma| \leq \text{OPT}$). Since there are exactly m iterations, we obtain the following theorem.

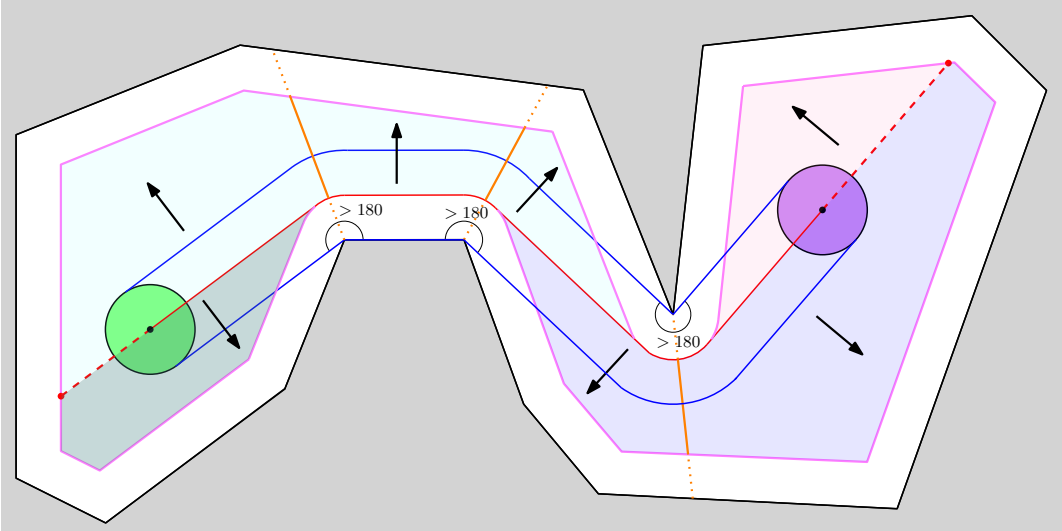


Figure 17: An iteration of the Exodus algorithm. The red curve is the extended geodesic $\hat{\gamma}$, and the blue curve is its trace. The free-space boundary $\partial\mathcal{F}$ is shown in pink. Each colored region represents a distinct pocket induced by $\hat{\gamma}$, and the orange rays are the angle bisectors generated at the corresponding reflex vertices of $\partial\mathcal{F}$. These bisectors subdivide each pocket into Exodus cells, within which a single direction vector is illustrated by the black arrows.

7.1 Partitioning the free space into cells

Consider a geodesic path $\gamma : [0,1] \to \mathcal{F}$ directed from $s \in S$ to $t \in T$. Recall that the edges of γ are either straight line edges or circular arcs of

the unit disk. Because $\omega = 3$, the positions s, t are not on $\partial\mathcal{F}$, and therefore they are the endpoints of straight line segment edges of γ. More precisely, γ is a sequence of edges $s_1, c_1, s_2, c_2, \ldots, s_k$, where $s_1, \ldots, s_k$ are straight line segments, and $c_1, \ldots, c_{k-1}$ are circular arcs.

We define the ***extension*** of the path γ, denoted $\hat{\gamma}$, to be the path obtained by extending the first edge of γ in the backwards direction and the last edge of γ in the forward direction, until both edges hit $\partial\mathcal{F}$. The extended path $\hat{\gamma}$ is also a geodesic path in $\mathcal{F}$, and it consists of straight line edges or circular arcs. Moreover, every circular arc of $\hat{\gamma}$ corresponds to a reflex vertex of $\mathcal{W}$. More precisely, for every circular arc of $\hat{\gamma}$ there exists a reflex vertex $v \in \mathcal{W}$ such that this arc is on the boundary of the closed unit disk centered at v, and therefore this arc is on the boundary of $\mathcal{F}$. We therefore conclude that $\hat{\gamma}$ splits $\mathcal{F}$ into two regions, one on either side of $\hat{\gamma}$. We denote these two regions by $\mathcal{F}_1(\gamma)$ and $\mathcal{F}_2(\gamma)$. Note that $\mathcal{F}_1(\gamma)$ and $\mathcal{F}_2(\gamma)$ are not necessarily connected regions.

A ***pocket*** of $\mathcal{F}_1(\gamma)$ is defined to be a maximally connected sub-region of $\mathcal{F}_1(\gamma)$, whose boundary consists of a portion of $\partial\mathcal{F}_1(\gamma)$ and a portion of $\hat{\gamma}$. Note that the points where these two portions meet must be circular arc edges of γ, that lie on circular arc edges of $\partial\mathcal{F}$ whose incident edges belong to $\partial\mathcal{F}_1(\gamma)$. Moreover, since each circular edge corresponds to a reflex vertex of $\mathcal{W}$, the portion of $\hat{\gamma}$ inside a single pocket must be a concave chain. The pockets of $\mathcal{F}_2(\gamma)$ are defined similarly.

Next, we divide each pocket into cells as follows. Consider a pocket P of $\mathcal{F}_1(\gamma)$ that does not contain s or t, and let $\gamma_P \subset \gamma$ be the portion of $\hat{\gamma}$ defining P. Let $c_1, s_1, c_2, s_2, \ldots, c_k$ be the sequence of edges of γ_P, where $s_1, \ldots, s_{k-1}$ are straight line segments, and $c_1, \ldots, c_k$ are circular arcs. Let $v_1, \ldots, v_k$ be the sequence of reflex vertices of $\mathcal{W}$ corresponding to $c_1, \ldots, c_k$. In addition, denote by s_0, s_k the straight line edges of $\hat{\gamma}$ prior to c_1 and following c_k, respectively. For every $1 \leq i \leq k$, we extend a ray originated at v_i towards P, bisecting the angle between s_{i-1} and s_i, until it hits $\partial\mathcal{F}$. These angle bisectors do not intersect, because the portion of $\hat{\gamma}$ in the pocket is concave. Because the polygon is simple, each sub bisector splits the pocket into two simple regions. Therefore, all these bisectors together split the pocket into a sequence of cells (see Figure 17), which we refer to as the *Exodus cells*, such that every two consecutive cells share a bisector. For a pocket P that contains either s or t, the sequence of edges of γ_P starts with a straight line edge instead of a circular arc edge. In any case, we define the Exodus cells for such pocket in the same way, by extending the bisectors from the corresponding reflex vertices.

Each Exodus cell C in a pocket P is defined by (at most) two consecutive reflex vertices v_i, v_{i+1} on the portion of $\hat{\gamma}$ that defines P, and it is bounded by:

1. the two angle bisectors corresponding to v_i, v_{i+1} (or a single angle bisector in case C contains one of the endpoints s or t),
2. a portion of $\partial\mathcal{F}$ that bounds P, between the angle bisectors, and
3. a portion of $\hat{\gamma}$, which consists of a portion of the circular arc edge c_i, followed by the straight line segment edge s_i, and then a portion the circular arc edge c_{i+1}. If C contains one of the endpoints s or t, then it is defined by a single reflex vertex (either the first or last in the chain), and contains only a single circular arc edge, and a straight line segment edge.

7.2 Clearing the geodesic path

We assign to each Exodus cell C a directional vector v_C, orthogonal to the single line segment edge of $\hat{\gamma}$ that bounds C, and pointing towards the interior of C (see Figure 17). Denote by $C_1, \ldots C_\ell$ the set of all exodus cells in all the pockets of $\mathcal{F}$. During the corridor-opening step, every robot whose center lies in C_i is translated by 2 units in the direction of v_{C_i}. Observe that since $\omega = 3$, for every $p \in S \cup T$ the disk $D_3(p)$ does not intersect the obstacle space $\mathcal{O}$. This means that ignoring collision with other robots, any robot positioned on p can move 2 units in any direction without colliding with obstacles. Therefore, we only need to show that this simultaneous movement by 2 units in direction v_{C_i} of all the robots in all the cells $C_1, \ldots C_\ell$, does not cause collisions between robots.

Lemma 15 *Let γ be a geodesic path from $s \in S$ to $t \in T$, and let $C_1, \ldots C_\ell$ be the set of Exodus cells constructed as above. The simultaneous movement by two units of all the robots (not including the robot positioned on s) in the direction v_{C_i} assigned to them according to the Exodus cell in which they lie, is collision-free.*

Proof. We will show that for any pair of robots A, B, the distance between their positions during the entire motion remains at least 2. Denote by $a_0, b_0 \in S \cup T$ the initial positions of A, B, respectively. Let C^A and C^B

be the Exodus cells in which they lie, and let v_{C^A} and v_{C^B} be the corresponding direction vectors. During the corridor-opening step we move them linearly:

$$a(t) \;=\; a_0 + 2t\, v_{C^A}, \qquad b(t) \;=\; b_0 + 2t\, v_{C^B} \qquad (t \in [0,1]).$$

Here $t \in [0,1]$ parameterizes the continuous motion from the initial configuration ($t = 0$) to the final configuration ($t = 1$). By the robots-separation assumption, $\|a_0 - b_0\| \geq 2$. It is therefore sufficient to show that the distance is always $\|a(t) - b(t)\| \geq 2$ for any of the possible configurations of C^A and C^B.

Recall that Exodus cells were obtained by refining the pockets created by the extended path $\hat{\gamma}$: each cell corresponds to a maximal subregion inside a pocket, in which $\hat{\gamma}$ contributes a single straight line segment edge, and hence a single well-defined inward normal direction. For any pair of cells C^A and C^B, one of the following four situations occurs.

Case 1: $C^A = C^B$. In this case the robots lie in the same cell, and by construction $v_{C^A} = v_{C^B} = v$. Thus

$$a(t) - b(t) \;=\; (a_0 + 2t\, v) - (b_0 + 2t\, v) \;=\; a_0 - b_0,$$

so the relative position of the two robots is constant throughout the motion. In particular,

$$\|a(t) - b(t)\| = \|a_0 - b_0\| \;\geq\; 2 \qquad \text{for all } t \in [0,1].$$

Case 2: C^A and C^B are on opposite sides of a straight line edge of $\hat{\gamma}$. Here C^A and C^B are separated by a common straight segment edge of $\hat{\gamma}$. Thus, the inward normals on the two sides are opposite, so the direction vectors satisfy $v_{C^A} = -v_{C^B}$. W.l.o.g., we choose coordinates so that this normal direction is the y-axis, and we can write

$$a(t) = (x_a,\, y_a + 2t), \qquad b(t) = (x_b,\, y_b - 2t).$$

The horizontal coordinates remain fixed, while the vertical coordinates move in opposite directions. Hence

$$\|a(t) - b(t)\|^2 = (x_a - x_b)^2 + (y_a - y_b + 4t)^2$$

is a strictly increasing function of t. In particular,

$$\|a(t) - b(t)\| \;>\; \|a_0 - b_0\| \;\geq\; 2 \qquad \text{for all } t \in (0,1],$$

and again the separation never drops below 2 during the motion.

Case 3: C^A and C^B share a common angle bisector on their boundary. Note that C^A and C^B may be either in the same pocket or in different pockets. Let v be the reflex vertex of $\mathcal{W}$ from which the bisector is extended in the construction of C^A and C^B. W.l.o.g., assume that v is at the origin, and that the bisector aligns with the y-axis. Also assume w.l.o.g. that C^A lies to the left of the bisector and C^B to the right of it. Then by the construction, since the y axis is bisecting the straight line edges corresponding to C^A and C^B, we have $v_{C^A} = (v_x, v_y)$ and $v_{C^B} = (-v_x, v_y)$; the y-components are equal, while the x-components are exactly opposite.

Notice that all the robots in C^A and C^B are moving farther away from the bisector chord that is shared by the cells. Therefore if A and B are not intersecting this chord, then they will not collide during their motion. Otherwise, if A and B intersect the bisector chord, then a must be to the left of the y-axis and b to right of it. In this case, the horizontal distance (distance between the x-coordinates) of $a(t)$ and $b(t)$ strictly increases with t, while their y-coordinates change in the same way:

$$a(t) = (x_a + 2tv_x,\, y_a + 2tv_y), \qquad b(t) = (x_b - 2tv_x,\, y_b + 2tv_y).$$

Hence

$$a(t) - b(t) = \big((x_a - x_b) + 2t(2v_x),\ (y_a - y_b)\big),$$

so the horizontal component of the difference vector grows in magnitude, while the vertical component remains constant. The Euclidean distance $\|a(t) - b(t)\|$ is therefore a strictly increasing function of t, and in particular

$$\|a(t) - b(t)\| \ > \ \|a_0 - b_0\| \ \geq \ 2 \qquad \text{for all } t \in (0, 1].$$

Case 4: C^A and C^B do not share a common boundary. Suppose the two robots A and B lie in Exodus cells that neither share a segment of $\hat{\gamma}$ nor an angle bisector. As we show in the previous cases, during the motion the robots do not cross the portion of $\hat{\gamma}$ defining their cell, and do not cross the angle bisectors. Therefore collisions between such robots are not possible.

These four cases cover all possible relative positions of two robots with respect to the extended path $\hat{\gamma}$, and the Exodus cells determined by it. In every case, the pairwise distance between robots is never smaller than its initial value (which is at least 2 by the robots-separation assumption) throughout the motion. Hence, the corridor-opening step is collision-free.

Note that the reverse motion is collision free as well, even if the target t is now occupied. This is because the robots-separation is 2 and therefore no robot is colliding with $D_1(t)$. □

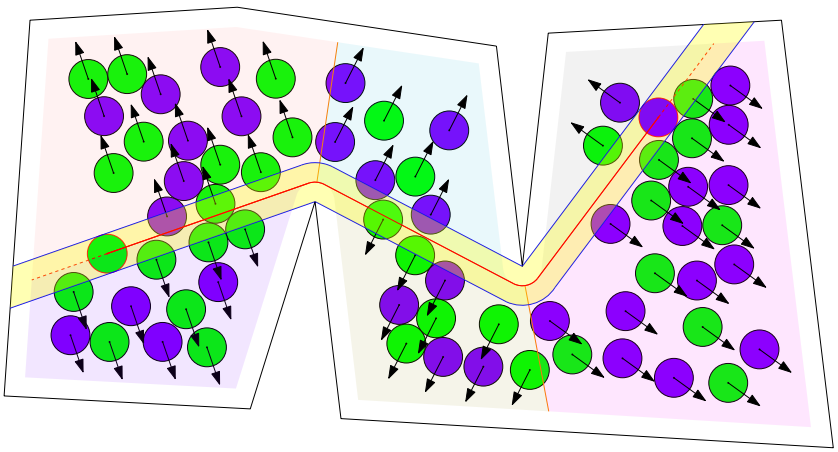


Figure 18: Illustration of one iteration of the *Exodus Algorithm*. The extended path $\hat{\gamma}$ (shown in red) partitions the free space into Exodus cells, each endowed with a consistently assigned direction vector. All robots whose centers lie in these cells (whether at the start or the target positions) are temporarily translated outward by exactly two units along their cell's assigned direction, thereby creating a collision-free corridor (shown in yellow) for the selected robot to advance toward its target. After the selected robot completes its motion, all displaced robots return to their original relative positions.

7.3 Running time and approximation analysis

Let m be the number of robots, and let OPT denote the total length of an optimal solution. We analyze the additional cost introduced by the global coordination mechanism of the Exodus Algorithm.

In each iteration, exactly one robot moves along its geodesic path γ_i toward its target, contributing exactly $|\gamma_i|$ to the total cost. The remaining $m-1$ robots are moving two units outwards from $\hat{\gamma}_i$ in order to clear its trace, and then move by the same distance after the robot moves from

s_i to t_i along γ_i. Thus each of the other $m-1$ robots executes an auxiliary motion of exactly 4 units per iteration. Since the algorithm performs exactly m iterations, the total auxiliary motion over all robots is $4(m-1)\cdot m = 4m^2 - 4m$.

The optimal-assignment path set Γ contributes at most OPT to the total length of the resulting motion plan, as the algorithm follows precisely the geodesic paths prescribed by the optimal assignment.

Consequently, the total length of all trajectories produced by the Exodus Algorithm is at most OPT $+ 4m^2 - 4m$.

We obtain the following theorem.

Theorem 16 *Given a simple polygonal workspace $\mathcal{W}$ with n edges, a set S of m starting positions and a set T of m target positions, such that $\rho = 2$ and $\omega = 3$, the Exodus algorithm computes in $\tilde{O}(m^3 + mn^2)$ time a collision-free motion plan of size at most OPT $+ 4m^2 - 4m$, where OPT is the length of an optimal solution.*

Proof. We begin by computing the shortest paths using a visibility-graph approach. The free space $\mathcal{F}$ has complexity $O(n)$ and can be constructed in $O(n \log^2 n)$ time [18]. We then build a visibility graph that includes all vertices of $\mathcal{F}$ as well as the m start and m target positions. The resulting generalized polygon has complexity $O(m+n)$, and its visibility graph can be constructed in $O(m^2 + n^2)$ time [10][1] . The graph contains $O(m+n)$ vertices and $O(m^2+n^2)$ edges. Running Dijkstra's algorithm from each s_i takes $O(m^2+n^2)$ time per execution. Since this procedure is carried out for all m robots, the total time for computing all start–target shortest paths is $O(m(m^2+n^2))$ time. Now to compute the optimal assignment Γ, we apply the Hungarian algorithm in $O(m^3)$ time [21].

One iteration of Exodus. In iteration i the algorithm focuses on γ_i and performs the following steps.

1. *Extending the corridor line.* The geodesic γ_i is extended at both endpoints along its tangent direction until it hits $\partial\mathcal{F}$, yielding the curve τ_i. This can be done in time linear in the complexity of γ_i, and hence in $O(n)$ time in the worst case.

[1] Unlike the purely polygonal visibility-graph setting, the free space $\mathcal{F}$ contains both line segments and circular arcs (arising from unit-disk offsets). But visibility-graph algorithms can handle such mixed arrangements, while still guaranteeing the same near-quadratic construction time.

2. *Path-induced regions, pockets, and Exodus cells.* The curve τ_i partitions $\mathcal{F}$ into two path-induced regions. Using the corresponding reflex vertices where τ_i changes direction, we construct the pockets along τ_i, and then subdivide each pocket by the angle bisectors (which compute in $O(n^2)$) into Exodus cells. This is a local refinement of $\mathcal{F}$ along τ_i, and its total complexity is $O(n)$. A simple polygonal decomposition together with point-location structure to it yields all the Exodus cells $O(n^2)$ time.

3. *Assigning displacement directions and moving robots.* Each cell C receives its displacement vector v_C. Given the cell decomposition, locating each robot center in its cell takes a query $O(\log n)$ expected time using the point-location structure we choose, so assigning all m robots to cells and applying the corresponding displacements costs $O(m \log n)$ time. By Lemma 15 In Section 7.2, this simultaneous two-unit displacement is collision-free.

Thus a single iteration runs in $O(n^2 + m \log n)$ time, and over all m iterations the execution phase contributes $O(mn^2 + m^2 \log n)$ time.

Total running time. Summing all contributions, the unlabeled version of Exodus runs in $O\left(mn^2 + m^2 \log n + m^3 + mn^2\right) = O\left(m^3 + mn^2\right)$ □

7.4 Labeled MRMP

In the labeled setting, where each start position is already associated with a specific target, we do not compute an optimal-assignment path set. Instead, we simply compute the geodesic paths between each start and its corresponding target using Wang's Euclidean shortest-path algorithm [33], and then apply the same iterative procedure described above. In this case, the running time becomes $O(m^2 \log n + mn^2)$, as no matching step is required. The motion of the robots during the corridor-opening iterations remains unchanged, and the size of the solution remains at most $\mathrm{OPT} + 4m^2 - 4m$.

8 Lower bounds

8.1 Lower bounds

In this chapter, we present lower bounds for the amount of obstacles-separation needed so that a solution to the MRMP problem always exists.

8.2 Lower Bound for Unlabeled MRMP

First, in the unlabeled case, we show that obstacles-separation of at least 1.5 is sometimes needed.

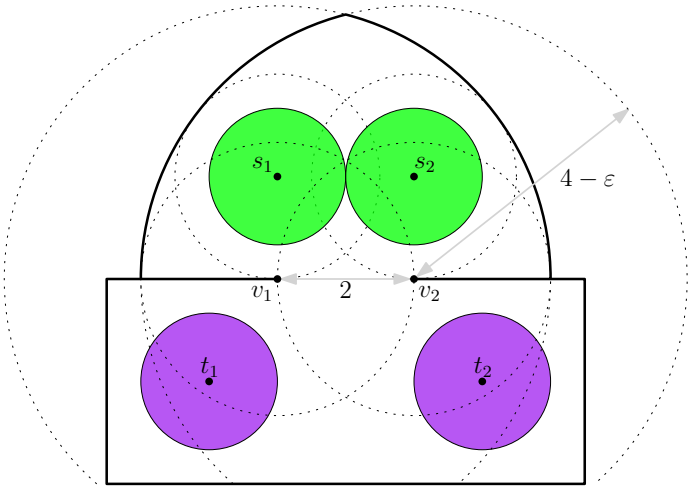


Figure 19: An instance of unlabeled MRMP with obstacles-separation of $1.5-\varepsilon$ for which no solution exists.

Lemma 17 *For $\omega < 1.5$ a solution does not always exist in the unlabeled variant, even if the free space consists of a single connected component containing two*

start and two target positions, and the bichromatic robots-separation is arbitrarily large.

Proof. Consider Figure 19. The positions s_1, s_2 are at distance exactly 2 from each other. The dotted disks centered at s_1 and s_2 are of radius $1.5 - \varepsilon$, for some small $\varepsilon > 0$. The point v_1 is directly below s_1 and v_2 is directly below s_2, so the distance between v_1 and v_2 is also exactly 2. The upper "room" is constructed using two arcs, the one on the right belongs to the disk of radius $4 - \varepsilon$ centered at v_2, and the one on the left belongs to a disk of radius $4 - \varepsilon$ centered at v_1. The positions t_1, t_2 lie in the bottom "room", below the opening between v_1 and v_2. Observe that the free space is a single connected region, as the narrow passage between v_1 and v_2 is of width 2. This implies that there exists a straight-line path within the free space connecting the two rooms of the polygon.

Now, in order for a robot A positioned on s_1 to reach the bottom room, the robot B lying on s_2 must step aside. Since the distance between v_1 and v_2 is exactly 2, when A passes through the narrow passage between v_1 and v_2, its boundary must touch both v_1 and v_2 at the same time. Consider the first time when the boundary of A touches v_1. At that time, A must be completely contained in $D_2(v_1)$. However, notice that the upper room is contained in a disk of radius $4 - \varepsilon$ around v_1, and therefore the robot B must intersect $D_2(v_1)$. If B is located to the right of A, it prevents A from reaching the passage. On the other hand, A and B cannot switch sides, i.e., B will always be to the right of A, because by symmetry A prevents B from passing above it. By symmetry, B cannot reach the passage as well, and therefore there is no solution to this instance for $\varepsilon > 0$.

Note that for $\varepsilon = 0$, the robots can rotate in a disk of radius 2 inside the upper room, so that when B reaches the top point in the disk, A can enter the narrow passage between v_1 and v_2. □

8.3 Lower Bound for Labeled MRMP

In the labeled case, we observe that obstacles-separation of at least 2 is sometimes necessary (for any value of ρ).

Lemma 18 *For $\omega < 2$ a solution does not always exist in the labeled variant, even if the free space consists of a single connected component containing two start and two target positions, and ρ is arbitrarily large.*

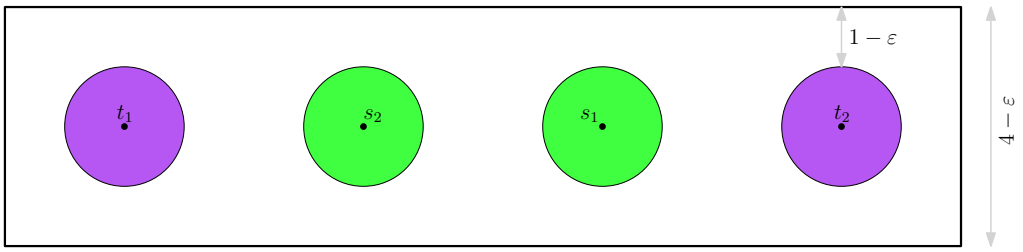


Figure 20: An instance of labeled MRMP with obstacles-separation of $2-\varepsilon$ for which no solution exists.

Proof. Consider Figure 20. The workspace is a long strip-shaped polygon of width $4-\varepsilon$ for some arbitrarily small $\varepsilon > 0$. The start and target points are aligned in the following order along the strip: t_1, s_2, s_1, t_2. The distance between any start/target position to the boundary is $2-\varepsilon$. In order for a robot A positioned on s_1 to move to t_1, it needs to switch sides with a robot B positioned on s_2. However, this is not possible: assume by contradiction that such a switch is possible, then at some point the x-coordinates of A and B should be equal, but this is not possible because the width of the strip is $4-\varepsilon$. □

We conclude that the best possible obstacles-separation (without further assumptions) that we can hope for is 1.5 in the unlabeled case, and 2 in the labeled case.

9 Conclusion

9.1 Summary and Conclusions

In this thesis, we addressed the foundational challenge of Unlabeled Multi-Robot Motion Planning (MRMP) in dense polygonal environments. By challenging the rigid geometric constraints of previous frameworks and introducing the concept of ε-overlap, we demonstrated that robots can safely navigate with significantly smaller separation bounds than previously thought possible.

The core motivation of this research was to find a better balance between the separation of the robots from each other and their separation from the obstacles. Previous approaches assumed a large separation between the robots ($\rho \geq 4$), which restricted the achievable robot density in the workspace. We demonstrated that by carefully managing a buffering layer around the obstacles (ω), the separation constraints between the robots themselves can be significantly relaxed. This methodology enables planning in high-density environments while ensuring that the total length of the computed paths remains within a constant factor of the optimal solution (OPT).

Most notably, we broke the long-standing $\omega = \sqrt{5}$ obstacles-separation barrier, positively resolving the conjecture by Solovey et al. [30], and established new, near-optimal weakly-monotone trade-offs (reaching up to $\omega \approx 1.354$). Furthermore, our introduction of the *Exodus algorithm* provided a novel global coordination strategy that achieves near-optimal robot density in simple polygons.

Beyond the theoretical advancements, we believe that the reduced separation bounds established in this work have direct implications for real-world robotic systems. By proving that robots can operate efficiently in significantly tighter spaces without sacrificing guaranteed feasibility or

near-optimality, this research provides the geometric and algorithmic foundations for more scalable and densely packed automated systems, from warehouse logistics to swarm robotics.

9.2 Technical Contributions

The main technical contribution of this work is a generalized planning framework that extends the optimality-driven approach of Solovey et al. [30] to significantly denser polygonal environments. Central to this framework is the introduction of an *overlap parameter* ε, which enables a formal distinction between two types of robot interactions:

Blocking positions: Configurations where the overlap with a moving robot's path strictly exceeds ε. These interactions fundamentally require a switching procedure to maintain feasibility.

Interrupting positions: Configurations with an overlap of at most ε. In these cases, stationary robots possess sufficient clearance to perform localized, synchronized displacements, dynamically clearing the path and returning to their original positions without demanding a global reconfiguration.

This geometric framework facilitates a smooth transition from strict monotone planning to a more flexible *weakly-monotone* coordination scheme. This approach rigorously preserves global correctness and constant-factor approximation guarantees for the total path length (OPT), while significantly relaxing the requisite separation assumptions.

Specifically, our analysis yields the following key results regarding the formal trade-offs between robots-separation (ρ) and obstacles-separation (ω):

The Monotone Planning Threshold. We resolved an open conjecture by proving that an obstacles-separation of $\omega = \sqrt{13 - 6\sqrt{3}} \approx 1.614$ is sufficient for strictly monotone planning when the inter-robot separation is $\rho = 4$. Furthermore, we demonstrated that below this exact threshold, strictly monotone motion plans may fundamentally fail to exist, establishing this value as an inherent geometric limit for monotone strategies.

Weakly-Monotone Trade-off Regimes. By adopting weakly-monotone coordination and assuming equal monochromatic and bichromatic separation (ρ), we established two highly improved trade-off regimes:

1. **Density-Optimized Regime:** $\rho = 2\frac{2}{3}$ and $\omega = 1\frac{2}{3}$. This regime aggressively pushes the required inter-robot separation closer to the theoretical geometric minimum.
2. **Clearance-Optimized Regime:** $\rho \approx 3.291$ and $\omega \approx 1.354$. This configuration pushes the necessary obstacles-separation toward the absolute optimal clearance baseline.

9.2.1 The Exodus Algorithm

To definitively approach the geometric lower bounds on robot density, we introduced the *Exodus algorithm*. This approach departs fundamentally from local clearing strategies: while our generalized framework relies on localized adjustments for robots within an overlap parameter ε, Exodus orchestrates a global, synchronized clearing motion. By temporarily displacing all stationary robots outward to carve a collision-free corridor, it enables efficient planning in even more densely packed environments.

This strategy seamlessly achieves a near-optimal robots-separation of $\rho = 2$, which precisely corresponds to the absolute geometric minimum for unit-disk robots. This milestone proves that coordinating at optimal robot density is tractable, albeit requiring a correspondingly higher obstacles-separation of $\omega = 3$. The Exodus framework is robustly applicable to both unlabeled and labeled MRMP, highlighting a qualitative paradigm shift from local avoidance to global macroscopic planning.

9.3 Lower Bounds and Limitations

To complement our algorithmic contributions, we established rigorous lower bounds defining the unavoidable geometric limits of the MRMP problem. This existential analysis sharply delineates the boundary between tractable planning and inherent mathematical infeasibility.

Obstacles-Separation Bounds. We demonstrated that certain minimum levels of obstacles-separation are strictly necessary for feasibility. In the

unlabeled setting, we proved that an obstacles-separation of at least $\omega \geq 1.5$ is sometimes unavoidable (note that this bound only works when no further assumptions are made on the robots-separation, i.e., when $\rho = 2$). In the labeled setting, this geometric requirement tightens further, where we proved that an obstacles-separation of $\omega \geq 2$ is sometimes necessary.

Limitations of Coordination Strategies. Furthermore, we identified the intrinsic limitations governing specific coordination patterns. We constructed geometric configurations where both monotone and weakly-monotone strategies definitively fail below certain separation thresholds:

- **Monotone Threshold:** Strategies strictly fail to yield a solution when $\omega < \sqrt{13 - 6\sqrt{3}} \approx 1.614$.
- **Weakly-Monotone Threshold:** Coordination inherently breaks down as the obstacles-separation drops below $\omega \approx 1.354$, even when the robots-separation is arbitrarily large.

These foundational results confirm that any future reductions in these separation parameters will mandate fundamentally novel coordination principles, moving entirely beyond the (weakly) monotone frameworks investigated in this thesis.

9.4 Discussion and Future Work

This thesis establishes the fundamental limits of monotone and weakly-monotone coordination strategies. While our constructive results significantly advance the boundaries of high-density multi-robot planning, they simultaneously illuminate the extreme frontiers where qualitatively new coordination mechanics are required.

9.4.1 Beyond Weakly-Monotone Strategies

A central conclusion of this work is the demonstration that weakly-monotone strategies cannot guarantee global feasibility once the obstacles-separation falls below $\omega \approx 1.354$. This realization poses a major theoretical challenge for the community:

Can we obtain better trade-offs between the obstacles-separation (ω) and the robots-separation (ρ) when using a fundamentally non-monotone strategy?

Determining whether a genuine "gap" exists between the capabilities of weakly-monotone coordination and the absolute existential feasibility of the problem remains an active and essential direction for the next generation of planning frameworks.

9.4.2 The Optimal Separation Frontier

For unit-disk robots, the absolute geometric lower bounds for any valid placement are a monochromatic-separation of 2, a bichromatic-separation of 0, and an obstacles-separation of 1.

The *Exodus algorithm* requires a robot-separation bound of $\rho = 2$. While this bound is tight for monochromatic separation, the bichromatic separation can potentially be reduced to 0. However, to safely achieve this extreme robot density, Exodus inherently necessitates an increased obstacles-separation of $\omega = 3$. In sharp contrast, our theoretical lower bounds establish that an obstacles-separation of at least $\omega \geq 1.5$ is sometimes strictly necessary to guarantee a valid motion plan for unlabeled MRMP.

This persistent gap between the constructive upper bounds of Exodus and our existential lower bounds raises compelling open questions:

What is the absolute minimum obstacles-separation for which we can remove the robots-separation assumption completely? Furthermore, is it mathematically possible to guarantee efficient planning when all separation parameters (μ, β, ω) approach their optimal geometric limits simultaneously?

Resolving these questions will likely require inventing new geometric tools capable of modeling swarms in extremely congested configurations, potentially utilizing the workspace boundaries themselves as continuous "sliding" surfaces.

9.4.3 Generalizations: Shapes and Environments

Building upon the rigorous insights gained from unit-disk robots, several natural, high-impact extensions arise:

- **Non-Circular Robots:** Investigating how these intricate separation trade-offs warp for square robots in orthogonal environments, or exploring rigorous bounding-circle approximations for general polygonal shapes.
- **Heterogeneous Groups:** Adapting the framework to maneuver swarms with varying radii, which will demand exponentially more complex hierarchical switching and clearing procedures.
- **High-Dimensional Workspaces:** Elevating "overlap-based" coordination into fully 3D environments, targeting applications such as drone swarms navigating highly constrained urban or subterranean spaces.

By resolving open geometric conjectures and establishing new, rigorously proven separation trade-offs, this thesis substantially advances our theoretical understanding of the critical boundary between tractable and intractable multi-robot planning. We hope that the geometric concepts and analytical tools formalized here will serve as a robust mathematical foundation for further progress toward fully optimal, swarm-level coordination in the densest of environments.

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תקציר

עבודה זו חוקרת את הבעיה של תכנון תנועה עבור מספר רב של רובוטים בצורת דיסק יחידה לא מתוייגים בתוך סביבה פוליגונלית. למרות שהבעיה הכללית נחשבת לקשה לפתרון מבחינה חישובית, ניתן למצוא פתרונות יעילים בזמן פולינומי תחת הנחות גיאומטריות מסוימות לגבי המרחק בין הרובוטים לבין עצמם ובינם לבין המכשולים בסביבה. עבודות קודמות הציעו פתרונות תחת תנאי הפרדה מקלים יחסית, מה שמעלה את הצורך באלגוריתמים המסוגלים לפעול בסביבות צפופות הרבה יותר.

בעבודה זו אנו מציגים אלגוריתמים המאפשרים יחסי גומלין משופרים בין חסמי הפרדת הרובוטים להפרדה מהמכשולים, המהווים שיפור משמעותי לעומת התוצאות הקיימות היום בחזית המחקר. אנו מציגים אלגוריתם קירוב בזמן פולינומי עבור משטרי הפרדה חדשים: (1) כאשר הפרדת הרובוטים היא $2\frac{2}{3}$ והפרדת המכשולים היא $1\frac{2}{3}$, (2) כאשר הפרדת הרובוטים היא בקירוב 3.291 והפרדת המכשולים היא בקירוב 1.354, וכן (3) פתרון מונוטוני הדורש הפרדה של 4 בין הרובוטים והפרדה של בקירוב 1.614 מהמכשולים. אנו מוכיחים כי תוכניות תנועה מונוטוניות עשויות שלא להתקיים מתחת להפרדה של 1.614 מהמכשולים, וכי תוכניות חלשות־מונוטוניות עשויות שלא להתקיים מתחת להפרדה של 1.354.

בנוסף, אנו מציגים את אלגוריתם אקסודוס, אסטרטגיית תיאום גלובלית המשיגה צפיפות רובוטים כמעט אופטימלית בתוך פוליגונים פשוטים. לסיום, אנו מוכיחים חסמים תחתונים המגדירים את המרחק המינימלי מהמכשולים ההכרחי לקיום פתרון עבור רובוטים לא מתוייגים. תוצאות אלו מאפיינות את הסף הגיאומטרי ההכרחי לקיום פתרון, ובכך הן משרטטות את הגבול שבין התחום הפתיר לבין המקרים שבהם הגיאומטריה מונעת קיום של מסלול. עבודה זו מגשרת על הפער שבין המורכבות התאורטית של הבעיה הכללית לבין התנאים הנדרשים ליישום מעשי של מערכות רובוטיות בסביבות עבודה עמוסות.



האוניברסיטה הפתוחה
המחלקה למתמטיקה ולמדעי המחשב

תכנון תנועה של רובוטים עם יחסי הפרדה משופרים

עבודת תזה זו הוגשה כחלק מהדרישות לקבלת תואר
"מוסמך למדעים" M.Sc. במדעי המחשב
באוניברסיטה הפתוחה
המחלקה למתמטיקה ומדעי המחשב

על־ידי
שלו גולדשטיין

בהנחיית ד"ר עומרית פילצר

מאי 2026